SCIENTIFIC COMMITTEE
TWENTY-FIRST REGULAR SESSION

Nuku'alofa, Tonga
13-21 August 2025

2025 Stock Assessment of Striped Marlin in the Southwest Pacific Ocean: Part II Data-moderate approach

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WCPFC-SC21-2025/SA-WP-07
19 July 2025

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2 Introduction

Assessments of Southwest Pacific Ocean (SWPO) striped marlin (*Kajikia audax*) have been challenging. As documented in the joint NOAA-SPC modeling meeting report (Ducharme-Barth et al., 2025), key issues identified by WCPFC SC20 included poor fits to size composition data and relative abundance indices, conflicts between different data sources, and difficulties in estimating model initial conditions.

Discussions at the 2025 SPC Pre-Assessment Workshop highlighted additional challenges related to the estimation of total population scale. The low biomass scale indicated by previous versions of the assessment appeared largely driven by the size composition from multiple fisheries. While estimation of low biomass scale is not inherently problematic, the resulting high ratio of fishing mortality (F) to natural mortality (M) that is maintained over multiple decades in the face of relative stability in catches and catch-per-unit-effort (CPUE) was noted as suspicious and flagged for further investigation.

The complexity of integrated age-structured models, while providing detailed population dynamics representation, can become problematic when fundamental data conflicts exist or when key biological parameters are uncertain. For SWPO striped marlin, these challenges are compounded by spatiotemporal heterogeneity in fleet coverage, potential non-representativeness in mixed-fleet composition data, and limitations in age data from opportunistic sampling. Changes to productivity assumptions (growth, natural mortality, maturity, and/or steepness) or selectivity patterns will impact biomass scaling through predictions of expected size composition data.

Scaling back model complexity to a data-moderate approach, like Bayesian surplus production models (BSPMs), offers analysts a simplified yet robust alternative for stock assessment when data limitations or conflicts present challenges for more complex models. These models have proven effective in recent pelagic fish stock assessments (Winker et al., 2018), and are routinely applied for WCPFC shark assessments (ISC, 2024; Neubauer et al., 2019, 2024). They facilitate a more tractable exploration of model assumptions by distilling productivity and fishing assumptions into a restricted parameter subset. The BSPM framework allows for explicit incorporation of parameter uncertainty through informative priors while maintaining computational efficiency and interpretability.

One advantage of the Bayesian approach is the use of prior information to propagate uncertainty in model assumptions. Biological simulation approaches can provide robust methods for parameterizing key population dynamics parameters (ISC, 2024; Pardo et al., 2016). For species with complex life histories like striped marlin, these simulation-based priors can incorporate uncertainty in growth, maturity, natural mortality, and reproductive parameters to derive realistic distributions for maximum intrinsic rate of increase and carrying capacity. Additionally, prior pushforward checks (Kim & Neubauer, 2025; Monnahan, 2024) can be used to further refine parameter distributions by identifying parameter combinations that produce clearly improbable outcomes.

This analysis presents a data-moderate approach for SWPO striped marlin that addresses some of the key technical challenges identified in previous assessments while providing a robust framework for stock status evaluation. The model incorporates biological uncertainty through simulation-based priors and includes approaches to address catch uncertainty and population depletion. It is important to note that this analysis complements the 2025 revised stock assessment report (Castillo-Jordan et al., 2025). While the relative simplicity of data-moderate approaches offers some advantages relative to more data-intensive approaches like fully-integrated age-structured stock assessment models, particularly when uncertainty in data or key assumptions is high, this comes at the cost of simplifying the model fishery and population dynamics. Over-simplification of key age-structured processes, or lack of spatiotemporal specificity could result in biases in data-moderate approaches. Taking multiple modeling approaches into account can give managers a holistic view on the likely status and trajectory of the stock.

3 Methods

3.1 Modelling framework

To address some of the issues raised in Section 2, a series of Bayesian state-space surplus production models (BSPMs) spanning the period 1952-2022 were developed following the Fletcher-Schaefer production model framework (Edwards, 2024; Winker et al., 2019). Development of the BSPM followed the approach of (ISC, 2024; Neubauer et al., 2019) and incorporated recent best practices for surplus production models (Kokkalis et al., 2024) and Bayesian workflows for stock assessment (Monnahan, 2024) as guides for model development, analysis, and presentation.

The models were implemented in the Stan probabilistic programming language (Team, 2024b) to take advantage of enhanced convergence diagnostics, greater efficiency in posterior sampling through the no-U-turn (NUTS) Hamiltonian Monte Carlo (HMC) algorithm (Betancourt & Girolami, 2013), and greater flexibility with model configuration and prior specification. Implementation in R using the *rstan* package (Team, 2024a) provides connection to an ecosystem of R packages for model diagnostics and validation: *bayesplot* (Gabry & Mahr, 2024) for posterior visualization and *loo* (Vehtari et al., 2024).

In their baseline configuration (diagnostic case), the models begin from an assumed unfished state in 1952 and incorporate process error in population dynamics, observation error in the relative abundance index and catch time series, and uncertainty in biological parameters. Population scale or carrying capacity $\log(K)$ is given in the units of the catch which is total numbers. The models fit to a single time series of relative abundance (e.g., standardized CPUE index) at a time. Rather than subtracting the catch directly, multiplicative survival after fishing mortality was applied. Fishing mortality was estimated internally by fitting to a time series of observed catches. Computationally, this was necessary to avoid placing hard constraints on the posterior parameter space which can cause convergence issues for the HMC sampling algorithm. Additionally, rather than directly estimating the independent fishing mortality values required to produce the catch, fishing mortality was derived from an estimated time-varying catchability trend and a time series of observed longline fishing effort. The primary rationale for this modelling choice being that it is easier to develop a defensible prior for catchability rather than an arbitrary prior for fishing mortality, given that previous work (ISC, 2024) showed a strong correlation between estimated population scale and the fishing mortality prior. Unless otherwise mentioned, all models used the same prior parameterizations (Table 2) based on the biological simulation filtering described in Section 3.5.

Using the diagnostic model[1] as a starting point, a number of one-off sensitivity analyses were conducted to explore key model assumptions or data questions. These are described in Section 3.6 and listed in Table 3. Based on the results of the sensitivity analyses a model ensemble was developed for the provision of stock

[1]The Stan code for the diagnostic model `bspm_estqsimple_softdep_fullmvprior_x0_sttgamma_flexsigmaC_OPT` is publicly available on GitHub.

status and management reference points. Full model development code and input data are available online at the following GitHub repository: https://github.com/N-DucharmeBarth-NOAA/2025-swpo-mls-bspm.

3.2 Input data

The BSPMs were fitted to catch and standardized CPUE data for SWPO striped marlin. Effort was used as an input to drive the time series of estimated fishing mortality via the estimated catchability trend.

Annual catch data (in numbers of individuals) spanning 1952-2022 were compiled from the 2024 assessment (Castillo-Jordan et al., 2024) sources and aggregated into total removals by year. The catch series (Figure 1) shows initially low removals in 1952-1953, followed by a high peak in 1954. Overall, however, catches have been relatively stable though catches in recent decades have shown a slight decline.

Annual longline effort (in numbers of hooks) matching the SWPO assessment domain, as specified in Castillo-Jordan et al. (2024), was extracted from publicly available WCPFC data. Longline effort in the assessment domain (Figure 1) increased gradually from zero in the early 1950s to ~100 million hooks per year in the early 1970s where it stabilized between 100-150 million hooks per year through the 1990s. Beginning in 2000 effort increased substantially to ~350 million hooks per year in 2003, and has fluctuated around that level for the last 20 years.

A number of indices were investigated either as one-off sensitivities or included in the structural uncertainty grid (Figure 2). These included the composite distant water fishing nation (DWFN) longline standardized CPUE index (1988-2022) from the diagnostic case of the 2024 assessment (Castillo-Jordan et al., 2024), an Australian longline index (Tremblay-Boyer & Williams, 2024), a New Zealand recreational sportfish index (Holdsworth, 2023), and three different observer data based indices (Neubauer et al., 2025). The Pacific Islands observer program indices contained information from New Caledonia, Fiji, Tonga, and French Polynesia observer programs. Three alternative indices were constructed either by including data from all programs together, including data from just the western programs (New Caledonia, Fiji, and Tonga) or including data just from French Polynesia. The DWFN, Australia and observer without French Polynesia indices show recent increases; the New Zealand and observer with only French Polynesia indices show recent declines; and the all observer program index shows no trend.

3.3 Population dynamics

The population dynamics follow a hybrid Fletcher-Schaefer surplus production model (Edwards, 2024; ISC, 2024) with state-space formulation where population depletion x_t (relative to carrying capacity K) evolves according to:

$$x_t = \left(x_{t-1} + \text{surplus production}_{t-1}\right) \times \epsilon_t \times \exp(-F_{t-1})$$

where surplus production is defined as:

$$\text{surplus production}_{t-1} = \begin{cases} R_{Max} \cdot x_{t-1} \left(1 - \frac{x_{t-1}}{h}\right), & x_{t-1} \leq D_{MSY} \\ \gamma \cdot m \cdot x_{t-1} \left(1 - x_{t-1}^{n-1}\right), & x_{t-1} > D_{MSY} \end{cases}$$

Here, R_{Max} is the maximum intrinsic rate of increase, n is the shape parameter, F_{t-1} is the fishing mortality in year $t-1$, and ϵ_t represents multiplicative process error with $\epsilon_t = \exp(\delta_t - \sigma_P^2/2)$ and $\delta_t \sim N(0, \sigma_P)$. The intermediate parameters are: $D_{MSY} = (1/n)^{1/(n-1)}$, $h = 2D_{MSY}$, $m = R_{Max} \cdot h/4$, and $\gamma = n^{n/(n-1)}/(n-1)$.

Fishing mortality F_t is in turn defined as a function of fishing effort and time-varying catchability:

$$F_t = q_{eff} \times \exp(\delta_{q,t} - \sigma_{qdev}^2/2) \times \mathrm{E}_t$$

where q_{eff} is the base catchability coefficient, E_t is the fishing effort[2] in year t, and $\delta_{q,t}$ represents temporal variation in catchability effectiveness.

The catchability deviations $\delta_{q,t}$ follow a structured temporal process organized into discrete periods of length n_{step} years. In the diagnostic model $n_{step} = 1$ which allows for annual catchability deviates across 70[3] periods p. Catchability remains constant within periods p, and evolves between periods according to an AR(1) process:

$$\delta_{q,p} = \rho \times \delta_{q,p-1} + \epsilon_{q,p} \times \sigma_{qdev} \times \sqrt{1-\rho^2}$$

where ρ is the autocorrelation parameter, σ_{qdev} is the standard deviation of catchability innovations, and $\epsilon_{q,p} \sim N(0,1)$. The initial period is constrained such that $\delta_{q,1} = 0$. The term $\sqrt{1-\rho^2}$ normalizes the variance of $\delta_{q,p}$ to remain stationary at σ_{qdev}^2 regardless of the value of ρ.

For each year t, the corresponding period assignment is:

$$\text{period}(t) = \min\left(\left\lfloor \frac{t-1}{n_{step}} \right\rfloor + 1, n_{periods}\right)$$

and the catchability deviation for year t is assigned as $\delta_{q,t} = \delta_{q,\text{period}(t)}$. This fishing mortality parametrization is a simplification of the *catch-errors* formulation in MULTIFAN-CL (Fournier et al., 1998) and allows catchability to vary smoothly over time while maintaining temporal structure and avoiding overfitting to annual fluctuations in the effort-catch relationship.

Model initial conditions are defined for $t = 1$ as $x_t = x_0 \times \epsilon_t$ where x_0 is the estimated depletion relative to unfished at the start of the model period. The prior specified for x_0 in the diagnostic case assumes an unfished initial condition with some uncertainty (Table 2).

3.4 Observation model

The model fits to a standardized CPUE index using a lognormal likelihood:

$$I_t \sim \text{Lognormal}(\log(q \times x_t) - \frac{\sigma_{O,t}^2}{2}, \sigma_{O,t})$$

[2]When input in the model the raw input effort, defined as *hundred hooks* in the WCPFC database is divided by `1e6` to re-scale catchability so that it is numerically on the same scale as other estimated parameters. This improves convergence of the HMC sampling of the posterior distribution.

[3]There are 71 total periods in the model period 1952-2022 and fishing mortality is not estimated in the terminal year.

where catchability q is analytically derived assuming an uninformative uniform prior, and total observation error $\sigma_{O,t} = \sigma_{O,input} + \sigma_{O,add}$ combines fixed input uncertainty with an estimated additional error component.

3.4.1 Catch treatment

Population removals (catch) were calculated in each time step as:

$$\text{Removals}_{t-1} = \left(x_{t-1} + \text{surplus production}_{t-1}\right) \times \epsilon_t \times (1 - \exp(-F_{t-1})) \times \exp(\log K)$$

As described in Section 3.3, instantaneous fishing mortality F_t was transformed to a discrete proportion of the population that survived fishing, survivorship, through the exponential decay function $\exp(-F_{t-1})$. Removals are simply calculated as the complementary proportion $(1 - \exp(-F_{t-1}))$ of the total production (including surplus production and process error) that are removed due to fishing and converted to absolute units, numbers, through $\exp(\log K)$.

The model fits to the observed total catch in numbers using a Student's t-distribution likelihood:

$$\log(Removals_{obs,t}) \sim \text{Student-t}\left(\nu_c, \log(Removals_t) - \frac{\sigma^2_{C,t}}{2}, \sigma_{C,t}\right)$$

where the degrees of freedom parameter ν_c controls the tail behavior[4], and the observation error $\sigma_{C,t}$ allows for time-varying uncertainty in catch observations. However, in the diagnostic model configuration $\sigma_{C,t} = 0.2$ for all t.

3.5 Prior development

The baseline prior distributions for leading estimated model parameters are specified in Table 2. A description of the derivation of these prior distributions follows.

Informed prior distributions for R_{Max}, and the Pella-Tomlinson shape parameter n were developed using numerical simulation based on biological knowledge. Informed prior distributions for $\log(K)$, and baseline catchability q_{eff} were developed using numerical simulation based on existing fishing data observations (e.g., catch and effort). Priors for other estimated leading model parameters (initial depletion x_0, AR1 correlation in catchability deviates ρ, variability in catchability deviates σ_{qdev}, process error σ_P, index observation error $\sigma_{O,add}$, and degrees of freedom of the catch likelihood ν_{catch}) were based on expert opinion and values observed in the literature. The naïve or input prior distributions were refined using a prior pushforward analysis.

3.5.1 Biological simulation framework

Informative priors for key population parameters were developed through biological simulation using Monte Carlo sampling from biologically plausible parameter distributions[5]. The simulation framework incorporated uncertainty and correlations in life history parameters:

[4] As $\nu_c \to \infty$ the likelihood equation converges to the lognormal distribution Lognormal $\left(\log(Removals_t) - \frac{\sigma^2_{C,t}}{2}, \sigma_{C,t}\right)$

[5] The R code for the simulation `01-01-define-rmax-prior.r` is publicly available on GitHub.

- **Growth parameters**: Francis parameterization with independent sampling
- **Natural mortality**: Reference mortality M_{ref} negatively correlated with maximum age (r = -0.3)
- **Maturity**: Length-based logistic function parameters
- **Reproduction**: Steepness parameter for stock-recruit relationship and sex ratio
- **Weight-length relationship**: Correlated allometric parameters (r = -0.5) with biological constraints
- **Fishery selectivity**: Length at first capture for depletion prior

Parameter combinations (Table 1) were filtered to ensure biological realism and included correlations between life history parameters reflecting biological trade-offs (e.g., shorter lived individuals having higher natural mortality rates).

3.5.1.1 Maximum intrinsic rate of increase (R_{Max}) prior

R_{Max} was calculated by numerically solving the Euler-Lotka equation using bounded optimization (nlminb function in R) to find the value of R_{Max} that satisfies the equation within convergence criteria. Starting values and bounds were set to ensure biologically realistic solutions. Formulation of the Euler-Lotka equation followed the approach implemented in openMSE (Hordyk et al., 2024; Stanley et al., 2009):

$\alpha \sum_{a=1}^{A_{Max}} l_a f_a \exp(-R_{Max} \times a) = 1$

where α is the slope at the origin of the stock-recruitment curve, l_a is the probability of survival to age a, and f_a is the fecundity (reproductive output) at age a. Additional technical detail can be found in Section 11.1 of the Technical Annex. Repeating this calculation across the 500,000 parameter combinations resulted in a prior distribution of R_{Max} conditioned on plausible biology and life-history characteristics. The resulting distribution was first filtered to values of $R_{Max} < 1.5$ representative of what the literature (Gravel et al., 2024; Hutchings et al., 2012) indicates is most likely for teleosts. A lognormal distribution was fit to the filtered distribution to derive input prior parameters for the Stan model.

3.5.1.2 Pella-Tomlinson shape parameter (n) prior

Similarly, a prior distribution for the Pella-Tomlinson shape parameter n was derived from the filtered R_{Max} values and associated biological parameter combinations through a two-step process. First, the inflection point of the production function was estimated using the relationship established by Fowler (1988):

$$\phi = 0.633 - 0.187 \times \log(T_G \times R_{Max})$$

where ϕ is the inflection point (representing the population level at which maximum sustainable yield occurs as a fraction of carrying capacity), T_G is the generation time calculated following Grant & Grant (1992) as $T_G = \sum_{a=1}^{A_{Max}} a \times l_a \times f_a / \sum_{a=1}^{A_{Max}} l_a \times f_a$, and R_{Max} is the maximum intrinsic rate of increase.

The Pella-Tomlinson shape parameter n was then derived from the inflection point through the mathematical relationship $\phi = (1/n)^{1/(n-1)}$, solved numerically using bounded optimization in R as done for R_{Max}. This transformation links the biologically-derived inflection point to the production function curvature, where $n = 2$ corresponds to the symmetric Schaefer model (e.g., MSY achieved at depletion $D = 0.5$). Lower values of n ($n < 2$) indicate more productive stocks that achieves MSY at $D < 0.5$ and higher values of n indicate less productive stocks with MSY occurring at $D > 0.5$. A lognormal distribution was fit to the distribution of

n to derive input prior parameters for the Stan model. Additionally, linking the shape parameter n to R_{Max} necessitated a multivariate prior structure so the correlation between the two parameters was estimated as well.

3.5.2 Fishing data simulation framework

Prior distributions for carrying capacity $\log(K)$ and initial catchability coefficient q_{eff} were derived using catch and effort data combined with the biologically-informed parameter distributions[6].

3.5.2.1 Carrying capacity prior ($\log(K)$)

The carrying capacity prior was constructed by assuming the maximum observed catch approximates maximum sustainable yield (MSY). This provides a conservative assumption for the upper limit of population scale[7] as larger MSY correlates positively with larger population scale. To account for uncertainty in the maximum catch $\sim$ MSY relationship (e.g., true MSY was above or below the maximum observed catch), candidate MSY values were sampled from a lognormal distribution centered on the maximum observed catch with moderate levels of uncertainty ($MSY \sim \text{Lognormal}(\log(\text{max catch}), 0.5)$).

For each combination of MSY, biologically-derived R_{Max} and shape parameter n, carrying capacity was calculated as:

$$K = \frac{MSY}{r \times \phi \times (1-\phi)^{n-1}}$$

where $\phi = (1/n)^{1/(n-1)}$ represents the inflection point of the production function. For the Fox model special case where $n = 1$, the relationship simplifies to $K = e \times MSY/r$. Non-finite values and extreme outliers ($\log(K) > 20$) were replaced through resampling from the valid parameter space to ensure numerical stability. A lognormal distribution was fit to the filtered distribution to derive input prior parameters for the Stan model.

3.5.2.2 Initial effort catchability prior (q_{eff})

Next, the initial catchability coefficient prior was developed using the relationship between CPUE and stock abundance:

$$q_{eff} = \frac{CPUE}{K}$$

where CPUE was calculated from early fishery data (first 10 years of catch and effort records) when the stock was assumed to be closer to carrying capacity. The prior distribution was constructed by randomly pairing CPUE values sampled from the early period with carrying capacity values from the $\log(K)$ prior distribution to generate a distribution of q_{eff} values incorporating both the uncertainty in population scale

[6] This is applied in the R script `03-06-new-logK-prior-pushforward-bspm_estqsimple_softdep_mvprior_x0.r` lines 64-92 which is publicly available on GitHub.

[7] The upper limit of population scale is typically more challenging to estimate than the lower population limit as population scales of infinite sizes can support any observed catches. However, there typically is a hard floor to population scale in that it must be sufficiently large to maintain a population that produces the time series of observed catches.

and the variability observed in early fishery CPUE. Again, a lognormal distribution was fit to the filtered distribution to derive input prior parameters for the Stan model.

3.5.3 Other priors

As mentioned previously, initial prior distributions for the remaining leading model parameters (initial depletion x_0, AR1 correlation in catchability deviates ρ, variability in catchability deviates σ_{qdev}, process error σ_P, index observation error $\sigma_{O,add}$, and degrees of freedom of the catch likelihood ν_{catch}) were specified either using expert opinion or literture.

Given that the limited large-scale industrial fishing existed in the SWPO prior to the expansion of the Japanese distant-water longline fleet in 1952, the population was assumed to start at an unfished condition with some uncertainty. Initial depletion x_0 was modeled with a lognormal distribution centered on 1. A lognormal prior was used for the standard deviation of the process error where the parameters were converted from the JABBA default prior for process error (Winker et al., 2018). The degrees of freedom nu_{catch} parameter assumed a gamma prior based on suggestions in the literature that it serves as a sensible default choice (Juárez & Steel, 2010). The prior for index observation error $\sigma_{O,add}$ assumed a half-normal distribution following ISC (2024) which provides sufficient support for a broad range of extra estimated observation error. For the autocorrelation coefficient in catchability deviations (ρ) and the standard deviation of catchability deviations (σ_{qdev}), initial priors were derived from data-filtering simulation results and specified as a bivariate distribution on the transformed scale: $\tanh^{-1}(\rho)$ and $\log(\sigma_{qdev})$.

3.5.4 Prior pushforward

Following Kim & Neubauer (2025), a prior pushforward analysis was conducted to refine the naïve prior distributions derived from the biological and fishing data simulation frameworks. Random parameter combinations drawn from the input prior distributions were passed through the diagnostic Stan model used for inference but with the likelihoods turned off (`fit_to_data = 0L`), generating simulated population trajectories and fishing dynamics under the model structure. This approach ensures that simulated trajectories are consistent with the actual model structure used for inference.

Parameter combinations were filtered through sequential criteria based on population viability and fishery realism (Figure 3):

- **Initial survival filter (** `iter_surve` **)**: Terminal year depletion $> 1\%$ (`mindep > 0.01`), minimum population size $>$ 15,000 individuals which matches recent observed catch levels (`minN > 15000`), and biologically reasonable shape parameter (`shape < 20`)
- **Fishing mortality filter (** `iter_f` **)**: Maximum fishing mortality < 2.5 (`maxF < 2.5`) and average fishing mortality < 0.8 (`avgF < 0.8`) to ensure realistic exploitation levels
- **Catch consistency filter (** `iter_catch` **)**: Simulated catches within realistic bounds relative to observed data: maximum catch between 0.5-5.0$\times$ observed maximum, minimum catch $> 0.5\times$ observed minimum, and average catch between 0.5-5.0$\times$ observed average
- **CPUE consistency filter (** `iter_cpue` **)**: Early period CPUE (first 10 years) between 0.5-5.0$\times$ observed early CPUE average to ensure realistic abundance levels during the initial fishery development

Parameter combinations passing all filtering stages (`iter_cpue`) were used to fit updated multivariate prior distributions using maximum likelihood estimation (Figure 4). Given that a number of parameters showed correlations following the prior pushforward, a 7-dimensional multivariate prior was developed for core parameters: $\log(K)$, $\log(R_{Max})$, $\log(n)$, $\log(x_0)$, $\log(q_{eff})$, $\tanh^{-1}(\rho)$ and $\log(\sigma_{qdev})$. Correlations with absolute values < 0.1 were reset to zero to avoid weak spurious correlations, retaining only meaningful parameter relationships. Though $log(x_0)$ was included in the multivariate prior, for convenience, it was treated as an unchanged univariate prior from the initial naïve prior distribution since the pushforward did not provide an update to this parameter.

Prior distributions for process error σ_P, index observation error $\sigma_{O,add}$, and degrees of freedom of the catch likelihood ν_{catch} were unchanged following the prior pushforward analysis and remained as univariate priors. Final prior distributions are listed in Table 2.

3.6 Description of one-off sensitivities

During the course of model development 38 one-off sensitivity models, organized into 12 groups, were investigated to explore alternative assumptions, model structures and sensitivities to data inputs. These models are listed in Table 3 and are described in further detail below.

3.6.1 Index

The diagnostic case model fits only to the DWFN index. One-off model sensitivities were conducted to explore how model estimates changed as a result of alternative indices:

- `0071` : Fits to the Australian longline index
- `0072` : Fits to the New Zealand recreational sportfish index
- `0073` : Fits to the longline observer data combining the programs from New Caledonia, Fiji, Tonga, and French Polynesia
- `0074` : Fits to the longline observer data excluding the French Polynesia program
- `0075` : Fits only to the longline observer data from the French Polynesia program

The French Polynesia observer program data was treated seperately as it showed differing trends from the other 3 observer programs (Neubauer et al., 2025). Additionally, given that the shorter indices (Australia and longline observer) provided little information about population scale, several other models were set-up to simultaneously fit them with the two longer indices (DWFN and New Zealand).

- `0129` : Fits to the DWFN & Australia index
- `0130` : Fits to the New Zealand & Australia index
- `0131` : Fits to the DWFN & Observer (New Caledonia, Fiji & Tonga) index
- `0132` : Fits to the New Zealand & Observer (New Caledonia, Fiji & Tonga) index

3.6.2 Catch observation error σ_C

The diagnostic case model assumes a constant catch observation error of $\sigma_C = 0.2$. Three one-off models explored alternative assumptions about catch uncertainty:

- `0076` : Time-varying σ_C following a power curve declining from 0.5 in early years to 0.2 in recent years, assuming improved catch reporting over time
- `0077` : Constant $\sigma_C = 0.1$, representing lower uncertainty in catch observations throughout the time series

- `0078` : Time-varying σ_C following a power curve declining from 0.5 in early years to 0.1 in recent years, combining both temporal improvement and lower overall uncertainty

3.6.3 Early data

Several models explored sensitivity to problematic early data years, particularly 1952 and 1954 which showed unusual catch or effort patterns. In particular the 1952 catch observation was anomalously low especially given the reported effort, and was identified to have high influence on model estimates. This could be a valid observation, as the Japanese longline fleet expanded into the south Pacific in 1952 they fished closer to the equator which tended to have lower striped marlin catch rates than in subsequent years when they fished between Australia and New Caledonia. Alternatively, this outlier point could be the result of erroneous reporting in either catch or effort. In either case, sensitivity to this data point was investigated:

- `0079` : Recalculates the low 1952 catch observation as the expected catch given the effort and average nominal CPUE from 1953-1957 (1952 catch is increased substantially).
- `0080` : Recalculates 1952 effort given the low 1952 catch and average nominal CPUE from 1953-1957 (1952 effort is decreased substantially)

Additionally, the large catch event in 1954 has been heavily scrutinized. This could be a valid observation as the Japanese longline fleet expended considerable effort during the time of year (3rd and 4th quarters of the year) striped marlin form spawning aggregations, in an area between Australia and New Caledonia where these aggregations form. Regardless, sensitivity to this observation was investigated in case it is a reporting error.

- `0081` : The 1954 catch was reduced by 50%
- `0082` : Combined alternative catch values for both 1952 and 1954
- `0083` : Combined alternative 1952 effort and 1954 catch values

These sensitivities address concerns about sensitivity to potentially anomalous observations in the early years of the fishery when reporting systems were less standardized.

3.6.4 Scaling priors

The diagnostic case model estimates a low population scale relative to the prior. Two models tested the impact of forcing the model toward higher population scale estimates:

- `0093` : Modified priors on log(K) and catchability (qeff) to favor higher carrying capacity and lower catchability estimates
- `0096` : Same as `0093` but combined with more conservative shape parameter prior (centered on 2)

These models explored whether alternative prior assumptions about population scale could shift model estimates.

3.6.5 Catch observation model

- `0110` : Replaced the default Student-t likelihood for catch observations with a lognormal error structure to test sensitivity to distributional assumptions about catch observation error

3.6.6 Temporal resolution of catchability deviates

The diagnostic case model estimates catchability deviates in single time steps (annual). Four models explored different temporal resolutions:

- `0111` : Catchability deviates fixed across 2-year periods (n_step = 2)
- `0112` : Catchability deviates fixed across 3-year periods (n_step = 3)

- `0113` : Catchability deviates fixed across 5-year periods (n_step = 5)
- `0114` : Catchability deviates fixed across 10-year periods (n_step = 10)

These sensitivities tested whether the temporal resolution of catchability changes affected model estimates, with longer periods resulting in a more parsimonious model but potentially diminishing the models ability to fit to data.

3.6.7 Catch reporting

Striped marlin are a commerically valuable non-target species, and as such it is assumed that if they are caught they are retained, and recorded in the logbook. However, given that they are a non-target species it is possible that they are not, or have not always been reported in the logbook and that under-reporting of catches or discarding could occur (e.g., given value of striped marlin and hold space required relative to target species). Three models explored potential systematic under-reporting of catches:

- `0115` : Doubled all observed catch values to test sensitivity to potential 50% under-reporting throughout the time series
- `0116` : Applied a linear multiplier increasing from 1x to 2x observed catch over time, representing improving catch reporting
- `0117` : Applied a linear multiplier decreasing from 2x to 1x observed catch over time, representing deteriorating catch reporting in recent years

3.6.8 Process error σ_P

The diagnostic case model estimated non-zero process error in the early model period, before the CPUE index began, and balanced process error with the catchability deviates in order to fit to the catch.

- `0118` : Restricted process error to be negligible ($\sigma_P \sim 0$) to evaluate the impact of environmental stochasticity on model estimates by forcing the population to follow deterministic surplus production dynamics

3.6.9 Start year

Given potentially anomalous catch observations in 1952 and 1954, along with uncertainty in the accuracy of early catch observations, four models explored sensitivity to the assumed initial conditions by starting the model in different years:

- `0119` : Model starts in 1955 with assumed initial depletion $x_0 = 0.9$, skipping the early data years

Three models were also tested which started the model in the same year as the CPUE index, and varied the level of initial depletion:

- `0120` : Model starts in 1988 with $x_0 = 0.5$
- `0121` : Model starts in 1988 with $x_0 = 0.7$
- `0122` : Model starts in 1988 with $x_0 = 0.9$

3.6.10 Inclusion of effort deviates

Early model development explored a more complex formulation of fishing mortality F_t where effort deviates were also used in addition to the catchability deviates to link effort to F_t:

$$F_t = q_{eff} \times \exp(\delta_{q,t} - \sigma^2_{qdev}/2) \times \mathrm{E}_t \times \exp(\delta_{e,t} - \sigma^2_{edev}/2)$$

This formulation of fishing mortality is closer to the full catch-errors with effort deviation approaches available in MULTIFAN-CL (Fournier et al., 1998). Two models tested whether estimating deviations from the effort time series improved model performance:

- `0124` : Estimated effort deviates with 2-year temporal resolution (n_step = 2)
- `0126` : Estimated effort deviates with 5-year temporal resolution (n_step = 5) for more constrained estimation

3.6.11 Direct estimation of fishing mortality

Rather than estimating fishing mortality indirectly through catchability and effort, two models directly estimated annual fishing mortality rates to see if the simpler formulation for fishing mortality could replicate the results of the diagnostic case model, or if the formulation of fishing mortality significantly impacted model estimates:

- `0127` : Direct F estimation with σ_F calibrated to match the variance in fishing mortality observed in the diagnostic case model
- `0128` : Direct F estimation with σ_F set to 0.25 times the diagnostic case variance, providing more constraint on fishing mortality variability in order to see how estimates of scale are impacted

3.6.12 Alternative R_{Max} prior

The diagnostic case model estimates a fairly productive stock with an R_{Max} that is considerably higher than an analogous species in the Atlantic Ocean, white marlin (*Kajikia albida*). The most recent ICCAT assessment (ICCAT, 2020), implemented using JABBA (Winker et al., 2018), estimated R_{Max} at 0.163 for white marlin. Sensitivity models assuming this lower level of productivity were developed to see the impact it would have on model estimates, particularly population scale.

- `0133` : Apply lower productivity white marlin Rmax prior
- `0134` : Apply lower productivity white marlin Rmax prior and apply a more conservative shape parameter prior (centered on 2)

3.7 Model Implementation

The models were implemented using the No-U-Turn Sampler (NUTS) in Stan. Each model was run using 5 independent Markov chains with randomly generated starting values to ensure robust exploration of the posterior parameter space. Each chain was sampled for 3,000 iterations, with the first 1,000 iterations discarded as warmup to allow for algorithm adaptation. To reduce posterior sample autocorrelation and storage requirements, every 10^{th} samples was retained from the post-warmup iterations, yielding a final sample size of 1,000 draws from the joint posterior distribution across all chains. The NUTS algorithm was configured with strict adaptation parameters to ensure reliable convergence. The adaptation delta was set to 0.99 to reduce the step size and minimize divergent transitions, while the maximum tree depth was increased to 12 to allow for longer trajectories during the dynamic sampling process.

Model convergence and performance were assessed using multiple diagnostic criteria recommended for Bayesian stock assessment workflows (Monnahan, 2024). The potential scale reduction factor ($\hat{R}$) was required to be less than 1.01 for all parameters, indicating convergence across chains. Effective sample sizes were required to exceed 500 to ensure adequate precision in posterior estimates. Additionally, posterior predictive checks were conducted to evaluate model adequacy through comparison of observed versus predicted index and catch values, while parameter identifiability was assessed through visual examination of posterior distributions and correlation structures among estimated parameters (Gabry et al., 2019).

3.8 Model diagnostics

Model performance was comprehensively evaluated using multiple diagnostic criteria recommended for Bayesian stock assessment workflows (Monnahan, 2024). The diagnostic framework included Stan model convergence criteria, data fits, posterior predictive checks, retrospective analysis, and hindcast cross-validation.

3.8.1 Convergence diagnostics

Conventional Stan model convergence diagnostics and thresholds were employed to identify whether posterior distributions were representative and unbiased (Monnahan, 2024). Models were considered to have 'converged' to a stable, unbiased posterior distribution if they satisfied the following criteria:

- **Potential scale reduction factor** ($\hat{R}$) less than 1.01 for all leading model parameters, indicating convergence across chains
- **Bulk effective sample size** (N_{eff}) greater than 500 for all leading model parameters, ensuring adequate precision in posterior estimates

- **No divergent transitions** during the NUTS sampling process, which can indicate problematic posterior geometry
- **Maximum tree depth** not exceeded during sampling, ensuring adequate exploration of parameter space

Additionally, trace plots and rank plots were examined to visually assess mixing and convergence behavior across chains (Gabry et al., 2019).

3.8.2 Leave-one-out cross-validation diagnostics

Model predictive performance and influential observations were assessed using leave-one-out cross-validation (LOO-CV) implemented through the *loo* package (Vehtari et al., 2017; Vehtari et al., 2024). LOO-CV provides an estimate of the expected log pointwise predictive density (ELPD) while identifying problematic observations that may unduly influence model predictions.

LOO-CV was also used as a factor to compare performance between models. Following the recommendations of Sivula et al. (2020), models were considered to have meaningfully different performance when the ELPD difference exceeded 4 units. For differences larger than 4 units, statistical significance was assessed by comparing the ELPD difference to twice its standard error, with differences greater than 2× the standard error considered statistically significant.

The effective number of parameters (p_loo) was examined as an additional diagnostic measure, with values substantially larger than the nominal number of model parameters indicating potential model misspecification. Catch and index observations were categorized as having low influence (Pareto-k $\leq$ 0.7), high influence (0.7 $<$ Pareto-k $\leq$ 1.0), or very high influence (Pareto-k $>$ 1.0). Observations with high or very-high influence were investigated to identify potential outliers, data entry errors, or systematic model misspecification that could compromise predictive performance.

3.8.3 Data fits

Model fits to the abundance index and catch time series were each quantified using normalized root-mean-squared error (NRMSE):

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}}$$

$$NRMSE = \frac{RMSE}{\sum_{i=1}^{n} y_i/n}$$

where y_i are the observed values and $\hat{y}_i$ are the model predictions. The average NRMSE across all posterior samples was calculated for each data component. Additionally, Probability Integral Transform (PIT) style residuals were calculated for the fit to the index.

3.8.4 Posterior predictive checks

Posterior predictive checks were conducted to evaluate model adequacy by comparing observed data with data simulated from the fitted model. For each posterior sample, synthetic datasets were generated using the estimated parameters, and agreement between observed and predicted datasets was assessed visually.

3.8.5 Retrospective analysis

Retrospective analysis was conducted for each model by sequentially peeling off a year from the terminal end of the fitted index and re-running the model. Data were removed for up to five years, from 2022 back to 2018. Estimates of x in the terminal year of each retrospective peel were compared to the corresponding estimate of x from the full model run to better understand any potential biases or uncertainty in terminal year estimates. The Mohn's ρ statistic (Mohn, 1999) was calculated and presented. This statistic measures the average relative difference between an estimated quantity from an assessment (e.g., depletion in final year) with a reduced time-series of information and the same quantity estimated from an assessment using the full time-series.

Additionally, following Kokkalis et al. (2024) the proportion of retrospective peels (or coverage) where the relative exploitation rate (U/U_{MSY}) and relative depletion (D/D_{MSY}) were inside the credible intervals of the full model run was calculated. This metric provides an additional perspective on retrospective consistency by evaluating whether the uncertainty estimates from the full model appropriately capture the range of estimates obtained when using reduced datasets.

3.8.6 Hindcast cross-validation

Hindcast cross-validation (Kell et al., 2021) was conducted for each index to determine model performance in predicting the observed CPUE I one-step-ahead into the future relative to a naïve predictor. Briefly, the 'model-free' approach to hindcast cross-validation was used, and used the same set of five retrospective peels described in Section 3.8.5.

The 'model-free' hindcast calculation is described using the model from the last peel $BSPM_{2017}$ as an example. This model fit to index data through 2017 but included catch through 2022. The model estimates of predicted CPUE in 2018 based on $BSPM_{2017}$ is the 'model-free' hindcast for 2018, $\hat{I}_{2018}$. The naïve prediction of CPUE in 2018 is simply the observed CPUE from 2017, $I_{2017} = \ddot{I}_{2018}$. The absolute scaled error (ASE) of the prediction is:

$ASE_{2018} = \frac{|I_{2018} - \hat{I}_{2018}|}{|I_{2018} - \ddot{I}_{2018}|}$

Repeating this calculation across all retrospective peels for years 2019-2022 and taking the average across ASE values gives the mean ASE or MASE for the model. An MASE value less than one indicates that the model has greater predictive skill than the naïve predictor.

3.8.7 Model-free hindcasts

Additional model-free hindcasts, similar to those described in Section 3.8.5, where the model was run for the full period 1952-2022 but without the last y years of index data. Hindcasts excluding the last 5, 10, 15, and 20 years of index data were conducted in order to evaluate the models ability to recapture the dynamics of the full model in the terminal year. Good hindcast performance over a long time horizon is an indication of a well-determined production function, and can give an idea of the reliability of the predictions from the full model.

3.9 Forecasts

Simple, status quo catch-based stochastic projections over a 10-year window are provided for each model. For each model, projected catch was taken as the average catch over the terminal 5 years (2018-2022). Process error in the projection period was randomly resampled from the estimated process error in the main model period.

4 Results

4.1 Diagnostic case

4.1.1 Model convergence and diagnostics

The diagnostic case model `0100` achieved satisfactory convergence based on standard Hamiltonian Monte Carlo diagnostics (Table 4). $\hat{R}$ values were below the 1.01 threshold and effective sample sizes exceeding the recommended minimum of 500 (100 samples per chain) for all parameters. No divergent transitions were observed across chains, and maximum tree depth was not exceeded during sampling, indicating successful exploration of the posterior space. Additionally, LOO-CV identified only two observations with Pareto-k $>$ 0.7 indicating the model structure and choice of likelihoods adequately matched the fitted data. Lastly, the effective number of estimated parameters p_loo was well below both the number of observations and the actual estimated parameter count indicating that model is likely to be reasonably well specified.

4.1.2 Model fit to data

4.1.2.1 Catch data fit

The diagnostic case model adequately captured the trend and magnitude of the observed catches (1952-2022) with the obvious exceptions of the 1952 and 1954 observations (Figure 5). Use of a catch likelihood, student-t, with heavier tails made the model more robust to these outlier observations, and accordingly estimated larger and smaller predicted catches, respectively, for these two observations. However, use of the student-t distribution along with an assumed 20% CV on the observation error did mean that catch observations were not fit exactly though the observed catches did tend to fall within the posterior predicted distribution (Figure 6). Given that SWPO striped marlin is a non-target species, some uncertainty in the predicted catches is considered to be appropriate.

4.1.2.2 CPUE index fit

Diagnostic case model fit to the standardized DWFN CPUE index (1988-2022) was evaluated using NRMSE and showed reasonable fits to the observed index given the high variability and observation error (Table 5). The model also successfully captured the general declining trend in standardized CPUE from 1988-2022 (Figure 7). PIT residuals did not show persistent positive or negative patterns, however they did indicate marginally better fit within the last decade of the index relative to the earlier portion(Figure 8). Posterior predictive checks confirmed that models replicated key features of the observed CPUE data (Figure 9).

4.1.3 Model validation

Overall retrospective bias was within the acceptable range proposed by (Hurtado-Ferro et al., 2015), however it did indicate a slight underestimate to the population recovery indicated by the DWFN index (Table 5; Figure 10). This is likely driven by the large observation in 2021 that is used by the full model to drag estimates up from the lower values observed in 2015-2019. Estimates of fishing mortality (U/U_{MSY}) and depletion (D/D_{MSY}) relative to MSY fell within the credible intervals of the full model run 100% of the time (Table 5). For hindcast cross-validation, the diagnostic model substantially exceeded the performance of the naïve predictor (Table 5; Figure 11). This provides support that models successfully identified and estimated a production function for the stock.

4.1.4 Parameter estimates

Following the guidance from Kim & Neubauer (2025), posterior distributions of parameter estimates were compared to the realized prior in addition to the naïve or input prior. The realized prior represents the effective prior distribution that results from passing naïve parameter combinations through the model structure with likelihoods turned off, revealing how model constraints and structural assumptions modify the input priors before data-based updating occurs (Kim & Neubauer, 2025). In the current analysis, there was little difference between the naïve and realized priors given the approach taken in developing the priors. Based on these comparisons the diagnostic model showed substantial posterior learning across several key leading parameters (Figure 12; Table 6), indicating that the data contained sufficient information to update parameter estimates from their prior distributions.

The carrying capacity $\log(K)$ exhibited substantial posterior learning, with the combination of index and catch data in the diagnostic case model strongly supporting lower population scales than suggested by either the naïve or realized priors. Similarly, the intrinsic rate of increase R_{Max} showed strong posterior updates, with the data favoring lower productivity than the biological simulation framework initially suggested. Inverse transform sampling to match the R_{Max} posterior distribution to the input biological simulation framework prior for R_{Max} indicates that the lower productivity is characterized by pronounced shifts toward lower steepness values h (Figure 13).

The median estimate of R_{Max} from the diagnostic case, 0.396 (Table 6), is considerably higher than estimates for an analogous species, white marlin (*Kajikia albida*), assessed using a JABBA surplus production model (ICCAT, 2020; Winker et al., 2018) in the Atlantic Ocean where $R_{Max} = 0.163$ (95% credible interval: 0.122-0.215). However, it is worth noting that the ICCAT assessment assumed a much more precise prior for R_{Max}, and that the 95% credible intervals of the diagnostic case model almost completely cover the ICCAT posterior distribution. Notably, the Atlantic white marlin JABBA model showed remarkable consistency with stock status and trend estimates from a fully integrated, length-structured model implemented in Stock Synthesis (ICCAT, 2020).

Shape and initial depletion parameters did not demonstrate notable posterior updates (Figure 12). The Pella-Tomlinson shape parameter n showed limited deviation from the prior, suggesting the data provided weak information about production function curvature. Initial depletion x_0 also remained close to the assumed unfished prior, indicating little information in the data to suggest large-scale fishing pressure prior to 1952.

The catchability and temporal variability parameters showed large posterior updates (Figure 12). The base catchability coefficient q_{eff} showed substantial learning, with strong negative correlations with $\log(K)$. This demonstrates a trade-off between population scale and catchability in order to fit the observed catches. If a more precise prior on q_{eff} were to be developed, this would be expected to set population scale and dramatically reduce uncertainty in model estimates. The autocorrelation parameter ρ and catchability variability σ_{qdev} both showed posterior updates, with the data informing a highly correlated temporal structure of catchability deviations.

Lastly, process error σ_P showed moderate posterior learning, while additional observation error $\sigma_{O,add}$ demonstrated substantial updates from the prior (Figure 12). This indicated that the model estimated additional uncertainty beyond the input index observation error, rather than using process error, to reconcile the observed CPUE variability. This results in a smoother year-on-year population trajectory. The degrees of freedom parameter ν_{catch} for the catch likelihood showed substantial updating (Figure 12), indicating the model preferred a Student-t distribution with very heavy tails in order to adequately account for the catch observations given the level of observation error.

4.1.5 Model-free hindcasts

Model-free hindcasting was conducted to evaluate the model's ability to predict population dynamics when fitted to progressively truncated index datasets. The diagnostic model was compared against retrospective peels excluding the final 5, 10, 15, and 20 years of index data (terminating in 2017, 2012, 2007, and 2002, respectively), with the model providing estimates through the full model period based on based only on the estimated production function and predicted catch.

The model demonstrated remarkably consistent performance across most hindcast scenarios (Figure 14) providing strong evidence of a well-determined production function. In general, the hindcast models successfully captured the general increasing trend in the standardized index after 2015. Model predictions for the DWFN CPUE index showed good coherence between the diagnostic model and all retrospective peels except the 15 year hindcast model, with overlapping 95% credible intervals throughout most of the time series. Estimates from the 15 year hindcast model appear to be strongly influenced by the series of higher observations (2003-2005) which cause it to underestimate the decline in the observed index.

Population dynamics estimates showed reasonable consistency across hindcast scenarios (Figure 15), with the notable exception of the 15 year hindcast model mendtioned in the previous paragraph. Depletion trajectories remained largely stable, with all models estimating similar patterns of decline from unfished conditions to minimum levels around the early 1990s, though predictions begin to diverge slightly in the recovery period post 2015. Population scale estimates also showed good agreement, though with increasing uncertainty for earlier time periods in longer hindcasts, as expected. The stock status metrics (D/D_{MSY} and F/F_{MSY}) showed fairly good consistency across most hindcast scenarios particularly for F/F_{MSY}. Additionally, given the The estimated trajectories for these key management metrics were nearly identical between the diagnostic model and all retrospective peels, suggesting that management advice derived from this model would have been stable regardless of when the assessment was conducted over the past two decades.

This consistency across multiple extended hindcast periods provides support that the underlying population dynamics are well described, given the available data. This gives confidence that projections made from the full model will likely capture the stock trajectory within the near to medium term.

4.2 One-off sensitivities

4.2.1 Index

4.2.2 Catch observation error σ_C

4.2.3 Early data

4.2.4 Scaling priors

4.2.5 Catch observation model

4.2.6 Temporal resolution of catchability deviates

4.2.7 Catch reporting

4.2.8 Process error σ_P

4.2.9 Start year

4.2.10 Inclusion of effort deviates

4.2.11 Direct estimation of fishing mortality

4.2.12 Alternative R_{Max} prior

4.3 Model ensemble

4.3.1 Ensemble diagnostics and fit

4.3.2 Population dynamics and stock status

4.3.3 Biological reference points

4.3.4 Future projections

5 Discussion

5.1 General remarks

The three BSPM variants provide a consistent picture of stock trajectory and current status. All models estimated similar depletion patterns: decline from unfished conditions in 1952 to minimum levels around the early 1990s, followed by gradual recovery through recent decades. This trajectory suggests the stock is currently recovering. While median estimates indicate the stock was not overfished or undergoing overfishing during the assessment period, the `0003-2024cpueFPrior` model suggests some recent risk of falling below these thresholds.

The BSPM approach addresses key limitations encountered in previous integrated assessments while providing robust stock status estimates. The primary advantage lies in condensing complex biological and fishery uncertainties into a manageable parameter set. Integrated assessments can be influenced by size composition data, which are sensitive to assumptions about growth, mortality, selectivity, and reproduction parameters. When these assumptions are uncertain or mis-specified, population scale estimates informed by size composition data can be biased. Production models circumvent this issue by aggregating these uncertainties into productivity parameters, allowing more efficient exploration of plausible population dynamics while maintaining biological realism through informed priors.

5.2 Challenges, limitations & key uncertainties

Several important limitations affect this analysis. The assumption of no age-specific population dynamics may be problematic when mortality or reproductive potential varies significantly by age. This simplification prevents incorporation of biological information beyond production function parameters and excludes explicit use of size composition data.

The aggregated nature of surplus production models can potentially bias MSY estimates when age-specific processes differ from production function assumptions. For species with complex life histories like striped marlin, managers should consider these limitations when applying MSY-based reference points derived from surplus production models.

Data limitations also constrain the analysis. Fitting to a single standardized abundance index limits evaluation of uncertainty in abundance trends. Multiple indices would enable model ensemble approaches to better characterize this uncertainty.

The incorporation of depletion information from New Zealand recreational size data through a 1988 depletion prior, while novel, represents a simplified approach. More sophisticated modeling frameworks like delay-difference or age-structured models could explicitly fit size composition time series in a likelihood-based framework, providing more elegant integration of this information.

Modeling assumptions introduce additional uncertainties. Current modeling assumes the same level of catch uncertainty across all time periods. Future versions could allow time-varying catch uncertainty (e.g., greater uncertainty in historical versus recent observations) or accommodate effort-driven removals similar to catch-errors with effort deviation approaches available in MULTIFAN-CL (Fournier et al., 1998).

Population scale estimates were highly uncertain and sensitive to model configuration choices, while depletion estimates were more precisely estimated. The choice of prior on σ_F was particularly influential. Although this prior was developed using prior pushforward analysis (Kim & Neubauer, 2025), it represents a key uncertainty. Similarly, uncertainty in biological parameters (growth, maturity, natural mortality) due to limited sample sizes contributed to broad initial priors for R_{Max}. While the posterior distribution for R_{Max} was substantially updated from the prior, reducing uncertainty in key biological productivity parameters could help reduce model uncertainty estimates and facilitate development of age-structured models.

5.3 Future stock assessment modeling considerations

Future assessment efforts should progressively build toward full age-structure through a systematic approach: delay-difference models, age-structured production models, and then fully integrated age-structured models. This development should occur within a Bayesian framework to inform parameter estimates where prior information exists or can be derived using prior pushforward approaches (Kim & Neubauer, 2025).

Developing Bayesian age-structured modeling approaches can provide more stable estimation of increased parameters due to the use of priors, explicitly account for parameter uncertainty rather than making fixed assumptions, potentially enable informed estimation of difficult-to-observe biological relationships (e.g., stock-recruit relationships), address BSPM limitations by explicitly considering age-structured processes and fishery selectivity, and estimate leading parameters with likelihood components for indices and size composition.

While simplified models offer advantages when data conflicts exist, over-simplification of age-structured processes could result in biases. Future modeling must balance complexity with data quality and conflicts. The transition should be guided by careful evaluation of whether increased model complexity is supported by available data and whether it improves the reliability of management advice.

6 Acknowledgements

The authors appreciate feedback received on earlier versions of this analysis, and are thankful to the following for their insights and thoughtful discussion which greatly improved the assessment: K. Kim (SPC), N. Davies (Te Takina Ltd.), S. Hoyle (Hoyle Consulting), R. Ahrens (PIFSC), and M. Nadon (PIFSC).

7 Declaration of generative AI use

A generative artificial intelligence (AI) assistant, Anthropic Claude Sonnet 4.0, was used to parse model code in the preparation of an initial draft of this report.

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9 Tables

Table 1: Biological parameter distributions and sources used in simulation framework

Parameter	Symbol	Distribution	Mean/Mode	CV/Range	Correlation	Source/Notes
Length at age 1	L_1	Lognormal	60 cm	0.2	Independent	(Ducharme-Barth et al., 2025)
Length at age 10	L_2	Lognormal	210 cm	0.2	Independent	(Ducharme-Barth et al., 2025)
Growth coefficient	k	Beta(6.5, 3.5)	~0.65	-	Independent	(Ducharme-Barth et al., 2025)
Length CV	cv_{len}	Uniform	-	[0.05, 0.25]	-	Growth variability
Maximum age	A_{Max}	Lognormal	15 years	0.2	ρ = -0.3 with M_{ref}	(Farley et al., 2021)
Reference mortality	M_{ref}	Lognormal	0.36 yr^{-1}	0.44	ρ = -0.3 with A_{Max}	(Hamel & Cope, 2022) ($5.4/A_{Max}$)
Maturity slope	a_{mat}	Normal	-20	0.2	-	Logistic steepness
Length at 50% maturity	L_{50}	Lognormal	184 cm	0.2	-	(Farley et al., 2021)
Weight coefficient	a_w	Lognormal	5.4×10^{-7}	0.05	ρ = -0.5 with b_w	(Castillo-Jordan et al., 2024)
Weight exponent	b_w	Normal	3.58	0.05	ρ = -0.5 with a_w	(Castillo-Jordan et al., 2024) Constrained: 150-350 kg at 300 cm
Steepness	h	Beta(3, 1.5)	~0.73	-	-	Censored to [0.2, 1.0]
Sex ratio (prop. female)	s	Normal	0.5	0.05	-	Constrained [0.01, 0.99]
Reproductive cycle	-	Fixed	1 year	-	-	Annual spawning
Reference ages	age_1, age_2	Fixed	0, 10 years	-	-	Francis parameterization

Table 2: Prior distributions for diagnostic model leading parameters.

Parameter	Distribution	Mean	SD	Description	Key Correlations
Core Population Parameters					
K	MV Lognormal	$\log(1, 110, 533)$	0.59	Carrying capacity (N)	$\rho(K, R_{Max})$ = -0.22; $\rho(K, q_{eff})$ = -0.241
R_{Max}	MV Lognormal	$\log(0.6206)$	0.58	Max rate of increase	$\rho(R_{Max}, n)$ = -0.881
n	MV Lognormal	$\log(1.16)$	0.54	Pella-Tomlinson shape parameter	
x_0	MV Lognormal	$\log(1)$	0.025	Initial depletion	(Independent)
q_{eff}	MV Lognormal	$\log(0.5019)$	0.95	Initial catchability	$\rho(q_{eff}, \sigma_{qdev}) = 0.559$; $\rho(q_{eff}, \rho) =$ 0.243
ρ	MV Transformed Normal	$\tanh^{-1}(0.981)$	0.62	Autocorrelation in catchability deviations	$\rho(\rho, \sigma_{qdev}) = 0.148$
σ_{qdev}	MV Lognormal	$\log(1.619)$	0.3	Catchability variability	
Process and Observation Error					
σ_P	Lognormal	$\log(0.0533)$	0.27	Process error	(Independent)
$\sigma_{O,add}$	Half-Normal	0	0.2	Additional estimated observation error	(Independent)
Catch Likelihood					
ν_{catch}	Gamma	Shape = 2	Rate = 0.1	Student-t degrees of freedom for catch	(Independent)

Notes:

Distributions with MV indicate that the parameter is part of the 7-dimensional multivariate prior.

Table 3: Model descriptions

Model	Group	Description
0100	Final ensemble	Diagnostic case model: DWFN index & baseline priors
0102	Final ensemble	New Zealand index & baseline priors
0105	Final ensemble	DWFN index & alternative shape prior
0107	Final ensemble	New Zealand index & alternative shape prior
0071	One-off: Index	Australia index
0072	One-off: Index	New Zealand index
0073	One-off: Index	Observer (all) index
0074	One-off: Index	Observer (New Caledonia, Fiji & Tonga) index
0075	One-off: Index	Observer (French Polynesia) index
0129	One-off: Index	DWFN & Australia index
0130	One-off: Index	New Zealand & Australia index
0131	One-off: Index	DWFN & Observer (New Caledonia, Fiji & Tonga) index
0132	One-off: Index	New Zealand & Observer (New Caledonia, Fiji & Tonga) index
0076	One-off: sigma_c	Time-varying sigma_c: Power curve from 0.5-0.2
0077	One-off: sigma_c	Constant sigma_c: 0.1
0078	One-off: sigma_c	Time-varying sigma_c: Power curve from 0.5-0.1
0079	One-off: Early data	Alternative 1952 catch
0080	One-off: Early data	Alternative 1952 effort
0081	One-off: Early data	Alternative 1954 catch
0082	One-off: Early data	Alternative 1952 & 1954 catch
0083	One-off: Early data	Alternative 1952 effort & 1954 catch
0093	One-off: Scaling priors	Force higher scale (logK & qeff)
0096	One-off: Scaling priors	Force higher scale (logK & qeff), alternative shape prior
0110	One-off: Catch obs model	Lognormal error structure
0111	One-off: qdev steps	n_step = 2
0112	One-off: qdev steps	n_step = 3
0113	One-off: qdev steps	n_step = 5
0114	One-off: qdev steps	n_step = 10
0115	One-off: Catch reporting	2x observed catch
0116	One-off: Catch reporting	Linear increase to 2x observed catch
0117	One-off: Catch reporting	Linear decrease from 2x observed catch
0118	One-off: Process error	Process error restricted to 0
0119	One-off: Start year	Start in 1955: x0 = 0.9
0120	One-off: Start year	Start in 1988: x0 = 0.5
0121	One-off: Start year	Start in 1988: x0 = 0.7
0122	One-off: Start year	Start in 1988: x0 = 0.9
0124	One-off: Est. effort devs	Estimate effort deviates & n_step = 2
0126	One-off: Est. effort devs	Estimate effort deviates & n_step = 5
0127	One-off: Direct F estimation	sigmaF to match diagnostic case
0128	One-off: Direct F estimation	0.25x diagnostic case sigmaF

Table 3: Model descriptions

Model	Group	Description
0133	One-off: Rmax prior	Apply lower productivity Rmax prior
0134	One-off: Rmax prior	Apply lower productivity Rmax prior and alternative shape prior

Table 4: Model diagnostics and fit statistics

Model	n par	n obs	Max $\hat{R}$	Min ESS	Divergent	Tree Depth	BFMI Issues	LOOIC	p LOO	ELPD LOO	Prop Pareto k
0100	150	114	1.008	788	0	0	0	45.4	32.7	-22.7	0.982
0102	150	118	1.008	688	0	0	0	64.9	32.9	-32.4	0.983
0105	150	114	1.008	677	0	0	0	46.1	33.0	-23.0	0.982
0107	150	118	1.007	523	0	0	0	66.1	33.3	-33.1	0.966
0071	150	95	1.007	792	0	0	0	-1.6	35.7	0.8	0.937
0072	150	118	1.014	610	0	0	0	65.3	33.7	-32.7	0.983
0073	150	92	1.008	738	0	0	0	-0.7	27.5	0.4	1.000
0074	150	92	1.008	760	0	0	0	30.2	28.9	-15.1	0.978
0075	150	90	1.036	230	0	0	0	58.3	29.7	-29.2	0.989
0129	150	139	1.014	666	0	0	0	24.7	36.5	-12.3	0.971
0130	150	143	1.012	760	0	0	0	56.2	39.3	-28.1	0.986
0131	150	136	1.005	710	0	0	0	35.9	35.5	-17.9	0.985
0132	150	140	1.011	783	0	0	0	61.7	36.7	-30.8	0.979
0076	150	114	1.008	644	0	0	0	49.8	19.2	-24.9	0.991
0077	150	114	1.013	693	0	0	0	32.0	57.3	-16.0	0.991
0078	150	114	1.010	779	0	0	0	28.6	26.9	-14.3	0.991
0079	150	114	1.010	754	0	0	0	19.5	33.6	-9.7	0.939
0080	150	114	1.010	773	0	0	0	20.9	34.3	-10.5	0.930
0081	150	114	1.006	748	0	0	0	45.8	32.1	-22.9	0.982
0082	150	114	1.009	748	0	0	0	21.3	33.4	-10.7	0.974
0083	150	114	1.008	537	0	0	0	21.5	33.3	-10.7	0.956
0093	150	114	1.006	749	0	0	0	52.4	35.3	-26.2	0.982
0096	150	114	1.007	746	1	0	0	57.0	35.1	-28.5	0.974
0110	149	114	1.008	729	0	0	0	1457.1	49.6	-728.5	0.711
0111	115	114	1.009	747	0	0	0	43.6	30.3	-21.8	0.974
0112	104	114	1.007	764	0	0	0	55.1	28.3	-27.5	0.974
0113	94	114	1.007	815	0	0	0	43.6	26.9	-21.8	0.982

Table 4: Model diagnostics and fit statistics

Model	n par	n obs	Max $\hat{R}$	Min ESS	Divergent	Tree Depth	BFMI Issues	LOOIC	p LOO	ELPD LOO	Prop Pareto k
0114	87	114	1.007	745	0	0	0	53.8	27.9	-26.9	0.991
0115	150	114	1.010	764	0	0	0	44.8	32.2	-22.4	0.991
0116	150	114	1.010	730	0	0	0	45.8	32.6	-22.9	0.982
0117	150	114	1.011	688	0	0	0	42.3	31.4	-21.2	0.982
0118	150	114	1.009	581	0	0	0	52.0	29.3	-26.0	0.991
0119	147	111	1.007	701	0	0	0	5.3	7.5	-2.6	0.977
0120	114	69	1.007	771	1	0	0	3.9	5.8	-2.0	0.971
0121	114	69	1.008	612	0	0	0	5.1	6.5	-2.6	0.971
0122	114	69	1.008	737	0	0	0	6.3	6.2	-3.1	0.971
0124	185	114	1.007	658	0	0	0	65.2	53.4	-32.6	0.860
0126	164	114	1.006	771	0	0	0	61.0	48.6	-30.5	0.904
0127	149	114	1.006	700	0	0	0	60.8	60.2	-30.4	0.535
0128	149	114	1.011	702	0	0	0	62.8	57.1	-31.4	0.544
0133	150	114	1.007	742	0	0	0	45.5	31.7	-22.8	0.982
0134	150	114	1.015	762	0	0	0	44.7	31.5	-22.4	0.982

Notes:

- n par: Number of parameters in the model
- n obs: Number of observations used in model fitting
- Max $\hat{R}$: Maximum potential scale reduction factor across all parameters (should be < 1.01)
- Min ESS: Minimum effective sample size across all parameters (should be > 500)
- Divergent: Number of divergent transitions during sampling (should be 0)
- Tree Depth: Whether maximum tree depth was exceeded during sampling

- BFMI Issues: Whether low Bayesian Fraction of Missing Information was detected
- LOOIC: Leave-one-out information criterion (lower is better)
- p LOO: Effective number of parameters estimated by LOO (should be < n par & n obs)
- ELPD LOO: Expected log pointwise predictive density from LOO (higher is better)
- Prop Pareto k: Proportion of Pareto k values <= 0.7 (should be high for reliable LOO estimates)

Table 5: Model performance and validation statistics

Model	Index RMSE	Catch RMSE	Mohn's ρ	MASE	Coverage D/D_{MSY}	Coverage U/U_{MSY}
0100	0.257	0.250	-0.087	0.587	100%	100%
0102	0.317	0.251	0.069	2.577	100%	100%
0105	0.258	0.250	-0.099	0.566	100%	100%
0107	0.317	0.256	0.087	2.579	100%	100%
0071	0.086	0.246	NA	NA	NA%	NA%
0072	0.316	0.250	NA	NA	NA%	NA%
0073	0.064	0.245	NA	NA	NA%	NA%
0074	0.188	0.244	NA	NA	NA%	NA%
0075	0.390	0.251	NA	NA	NA%	NA%
0129	0.189	0.248	NA	NA	NA%	NA%
0130	0.231	0.247	NA	NA	NA%	NA%
0131	0.230	0.251	NA	NA	NA%	NA%
0132	0.271	0.249	NA	NA	NA%	NA%
0076	0.261	0.335	NA	NA	NA%	NA%
0077	0.253	0.210	NA	NA	NA%	NA%
0078	0.256	0.307	NA	NA	NA%	NA%
0079	0.256	0.222	NA	NA	NA%	NA%
0080	0.256	0.223	NA	NA	NA%	NA%
0081	0.259	0.240	NA	NA	NA%	NA%
0082	0.256	0.221	NA	NA	NA%	NA%
0083	0.256	0.221	NA	NA	NA%	NA%
0093	0.275	0.247	NA	NA	NA%	NA%
0096	0.298	0.249	NA	NA	NA%	NA%
0110	0.250	0.201	NA	NA	NA%	NA%
0111	0.255	0.256	NA	NA	NA%	NA%
0112	0.260	0.297	NA	NA	NA%	NA%
0113	0.249	0.273	NA	NA	NA%	NA%
0114	0.239	0.316	NA	NA	NA%	NA%
0115	0.256	0.254	NA	NA	NA%	NA%
0116	0.258	0.228	NA	NA	NA%	NA%
0117	0.251	0.282	NA	NA	NA%	NA%
0118	0.281	0.258	NA	NA	NA%	NA%
0119	0.255	0.218	NA	NA	NA%	NA%
0120	0.272	0.170	NA	NA	NA%	NA%
0121	0.272	0.170	NA	NA	NA%	NA%
0122	0.279	0.171	NA	NA	NA%	NA%
0124	0.249	0.234	NA	NA	NA%	NA%
0126	0.245	0.238	NA	NA	NA%	NA%
0127	0.247	0.221	NA	NA	NA%	NA%

Table 5: Model performance and validation statistics

Model	Index RMSE	Catch RMSE	Mohn's ρ	MASE	Coverage D/D_{MSY}	Coverage U/U_{MSY}
0128	0.268	0.221	NA	NA	NA%	NA%
0133	0.260	0.252	NA	NA	NA%	NA%
0134	0.259	0.250	NA	NA	NA%	NA%

Notes:

- Index RMSE: Root mean squared error for fit to standardized CPUE index (lower is better)
- Catch RMSE: Root mean squared error for fit to catch data (lower is better)
- Mohn's ρ: Retrospective bias statistic (values close to 0 indicate low bias)
- MASE: Mean Absolute Scaled Error from hindcast cross-validation (< 1 indicates model outperforms naïve predictor)
- Coverage D/D_{MSY}: Percentage of retrospective peels where relative depletion estimates fall within full model credible intervals
- Coverage U/U_{MSY}: Percentage of retrospective peels where relative exploitation rate estimates fall within full model credible intervals

Table 6: Posterior estimates (median and 95% credible intervals) of the leading model parameters for models in the final ensemble.

Model	$\log(K)$	R_{Max}	n	x_0	q_{eff}	ρ	σ_{qdev}	σ_P	$\sigma_{O,add}$	ν_{catch}	σ_F
0100	12.786	0.396	1.353	1.000	2.227	0.993	1.495	0.058	0.084	3.797	—
	(12.323, 14.046)	(0.155, 0.866)	(0.500, 3.544)	(0.948, 1.053)	(0.602, 8.505)	(0.980, 0.997)	(1.013, 2.211)	(0.034, 0.094)	(0.029, 0.154)	(2.119, 7.667)	
0102	12.631	0.543	1.090	1.000	2.502	0.993	1.478	0.069	0.184	3.664	—
	(12.142, 14.378)	(0.185, 1.168)	(0.446, 3.042)	(0.952, 1.051)	(0.513, 9.714)	(0.979, 0.997)	(1.016, 2.210)	(0.038, 0.115)	(0.100, 0.291)	(1.967, 7.697)	
0105	12.751	0.322	2.913	1.002	2.356	0.993	1.547	0.059	0.083	3.747	—
	(12.198, 14.209)	(0.121, 0.722)	(1.172, 8.168)	(0.950, 1.045)	(0.596, 8.655)	(0.979, 0.997)	(1.041, 2.191)	(0.036, 0.101)	(0.029, 0.160)	(2.000, 7.582)	
0107	12.571	0.438	2.351	1.000	2.673	0.993	1.503	0.067	0.184	3.746	—
	(12.015, 14.428)	(0.153, 1.008)	(0.899, 6.335)	(0.954, 1.052)	(0.565, 9.271)	(0.980, 0.997)	(1.006, 2.147)	(0.038, 0.112)	(0.103, 0.291)	(1.988, 7.349)	

10 Figures

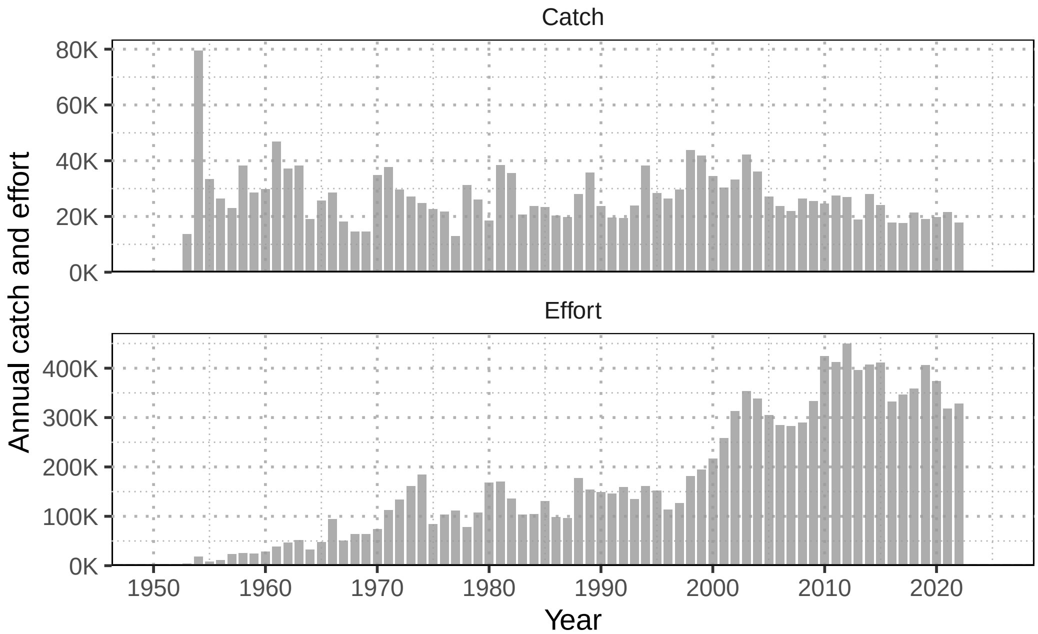

Figure 1: Annual catch (numbers; individuals) of striped marlin and nominal longline effort (hooks fished; thousands) in the Southwest Pacific Ocean (1952-2022).

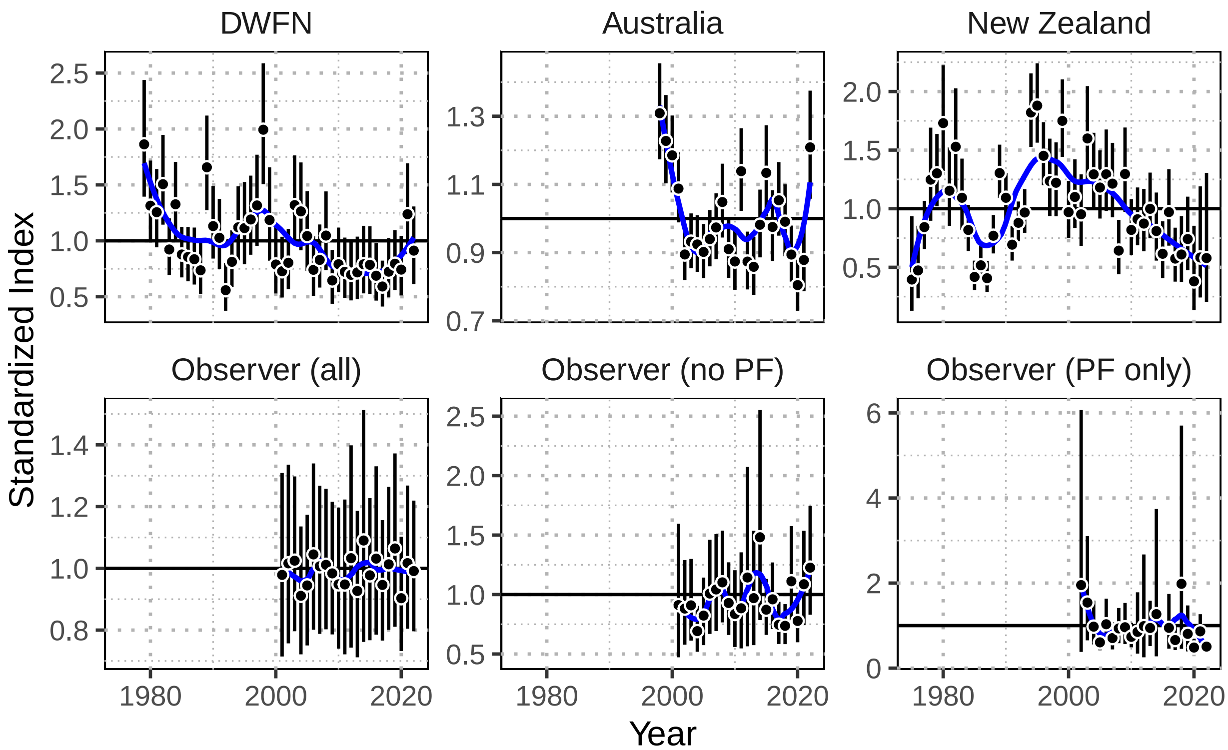

Figure 2: Standardized index (black points), and observation error (black bars) for alternative model indices. The trend in the indices is shown in blue using a loess-smooth.

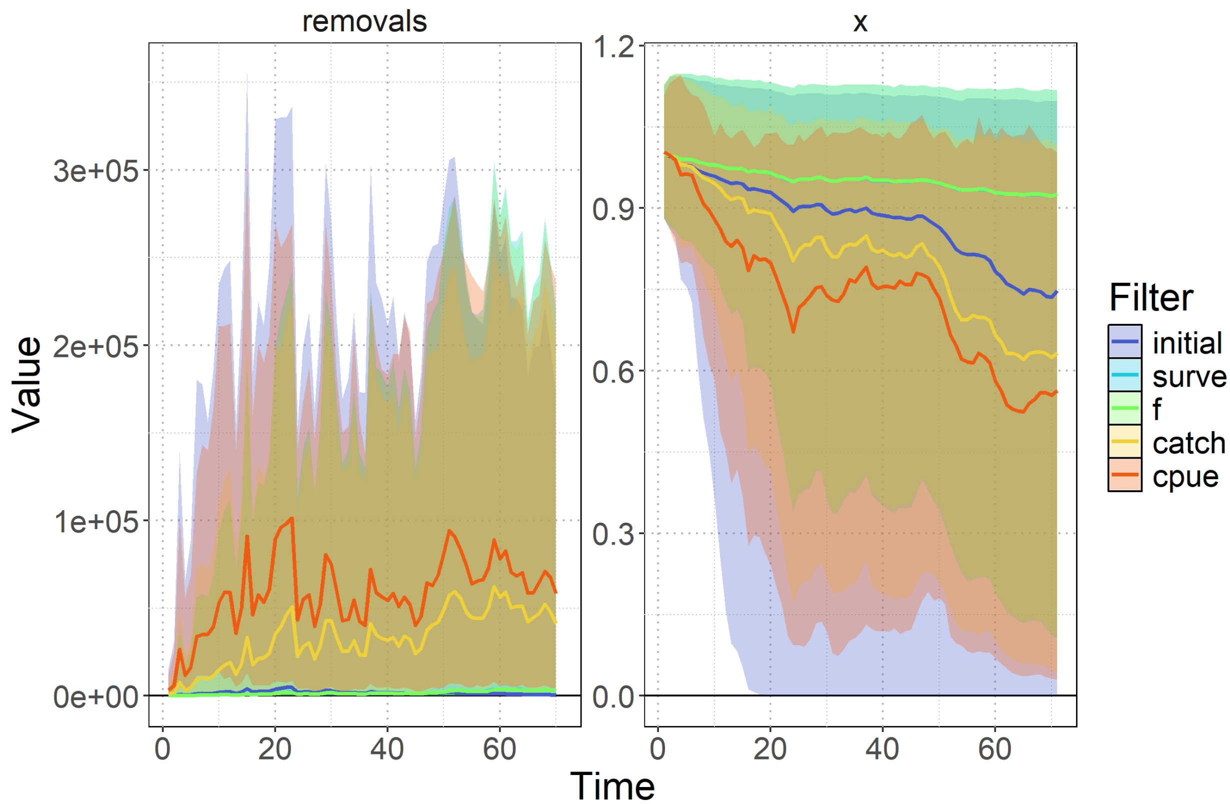

Figure 3: Prior pushforward check showing simulated removals and population depletion trajectories under different filtering scenarios (median: colored lines) with 95% quantile of trajectories (shaded polygon).

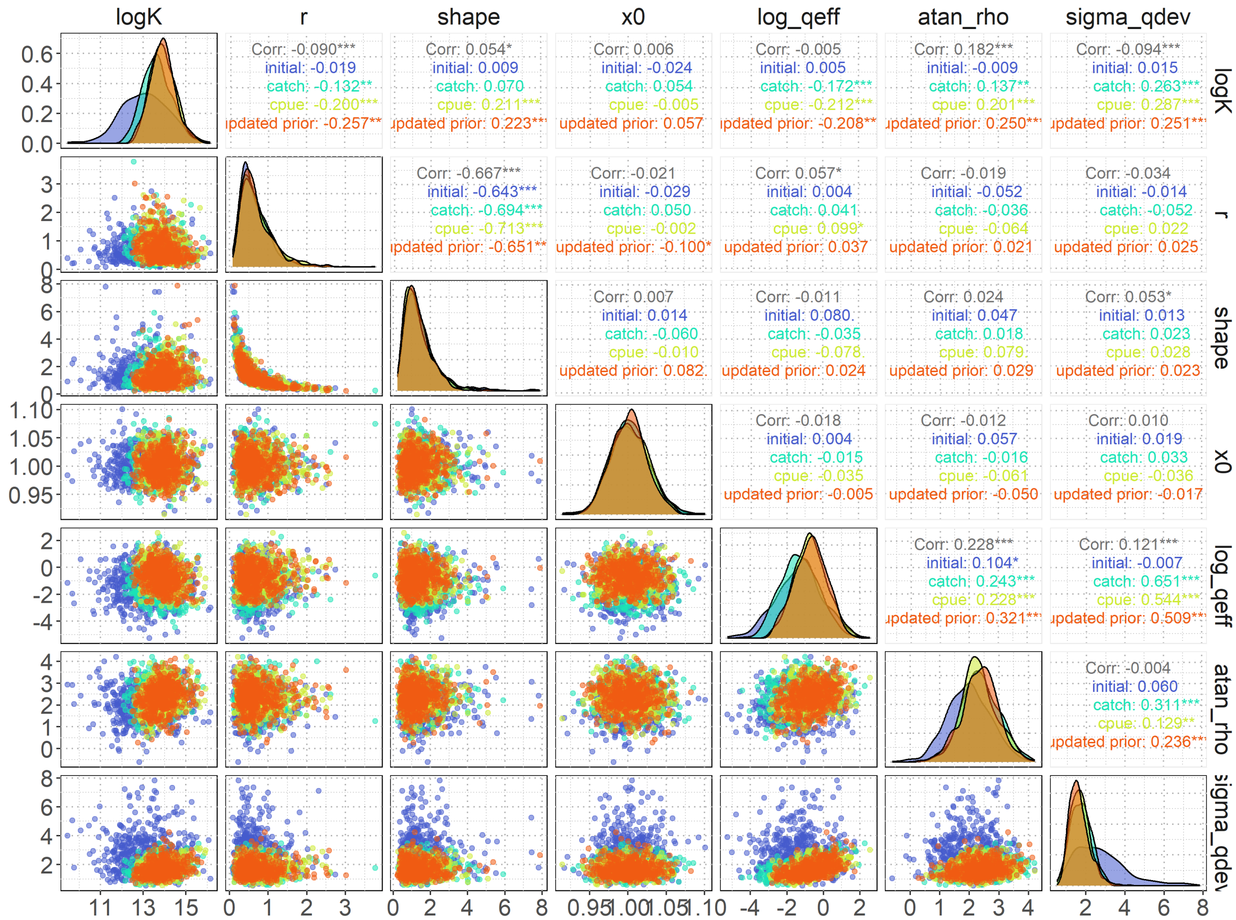

Figure 4: Prior pushforward diagnostic pairs plot showing parameter correlations and marginal distributions for different filtering scenarios. The plot compares initial priors (blue), catch-filtered priors (cyan), cpue-filtered priors (lime green), and the final updated priors following fitting to the cpue-filtered distribution (orange) across key model parameters (logK, r, shape, x0, log_qeff, atan_rho, sigma_qdev). Diagonal panels show marginal density distributions with overlaid curves for each filtering scenario. Off-diagonal scatter plots display pairwise parameter relationships colored by filtering type. Correlation coefficients in the upper triangle quantify linear relationships between adjacent distributions, with statistical significance indicated by asterisks.

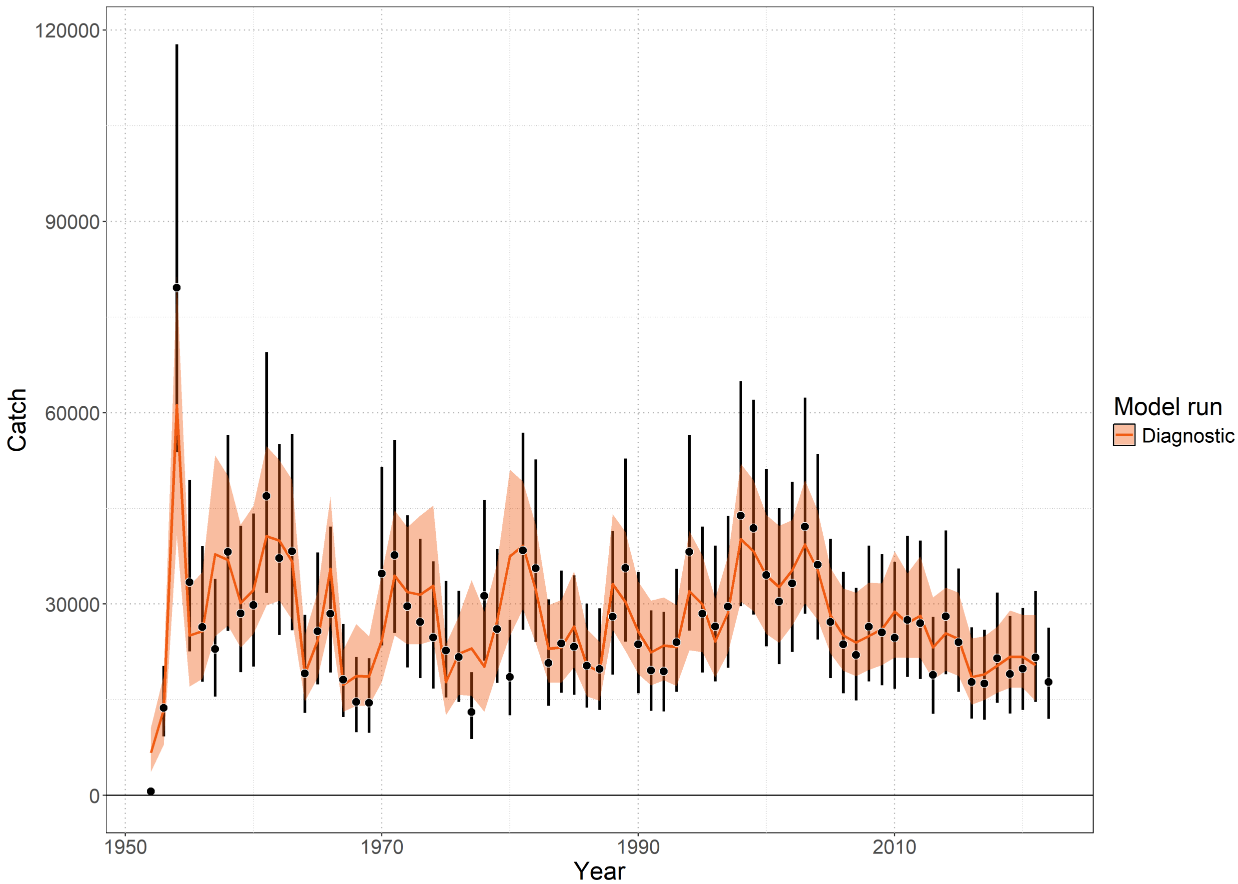

Figure 5: Model fit to catch data showing observed catch (black points with 95% error bars) and model-predicted catch (colored lines and shaded ribbons representing 95% credible intervals).

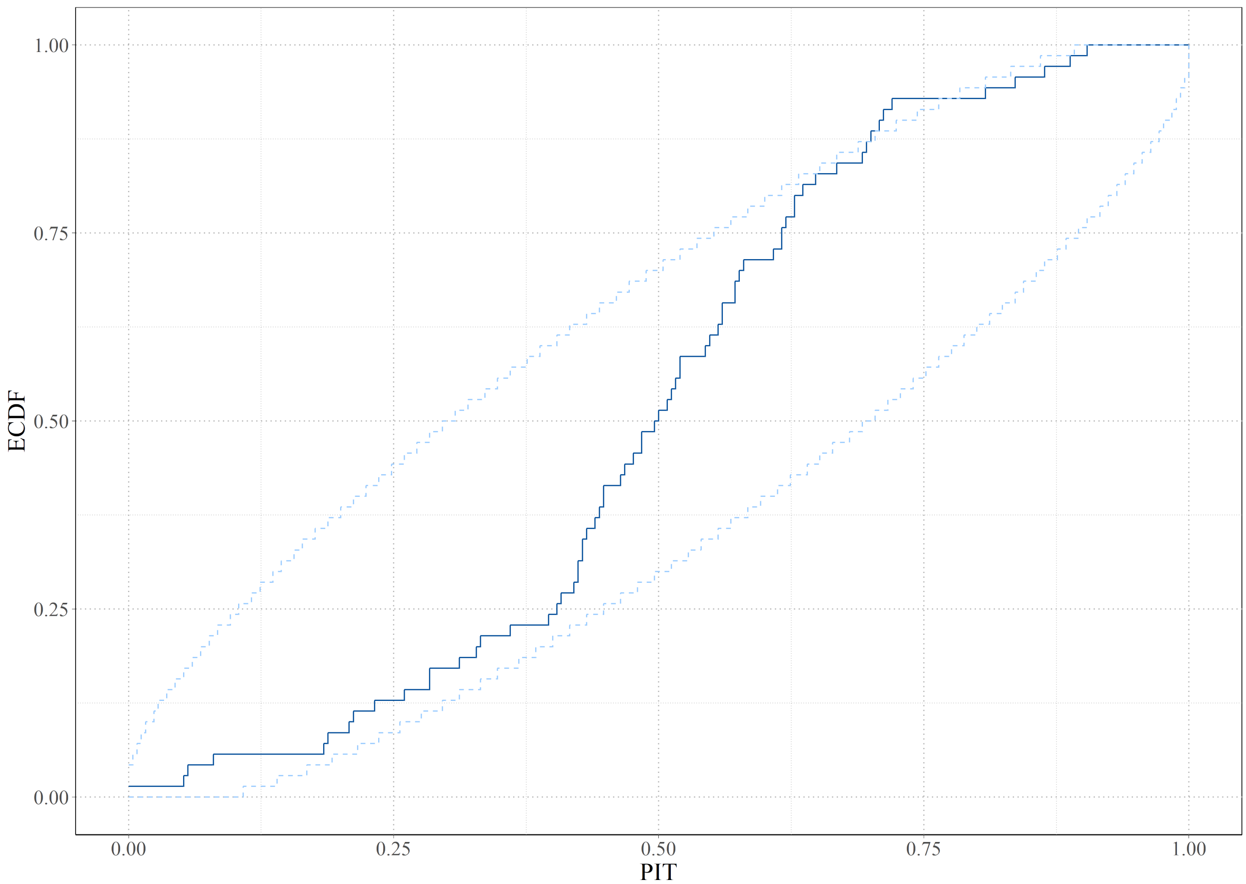

Figure 6: Posterior predictive check for catch data using probability integral transform empirical cumulative distribution function (PIT-ECDF). The solid blue line shows the empirical CDF of PIT values calculated from observed catch data relative to posterior predictive distributions. Light blue dashed lines indicate 99% simultaneous confidence bands under the assumption of correct model specification. The solid blue line should be contained within the 99% simultaneous confidence bands.

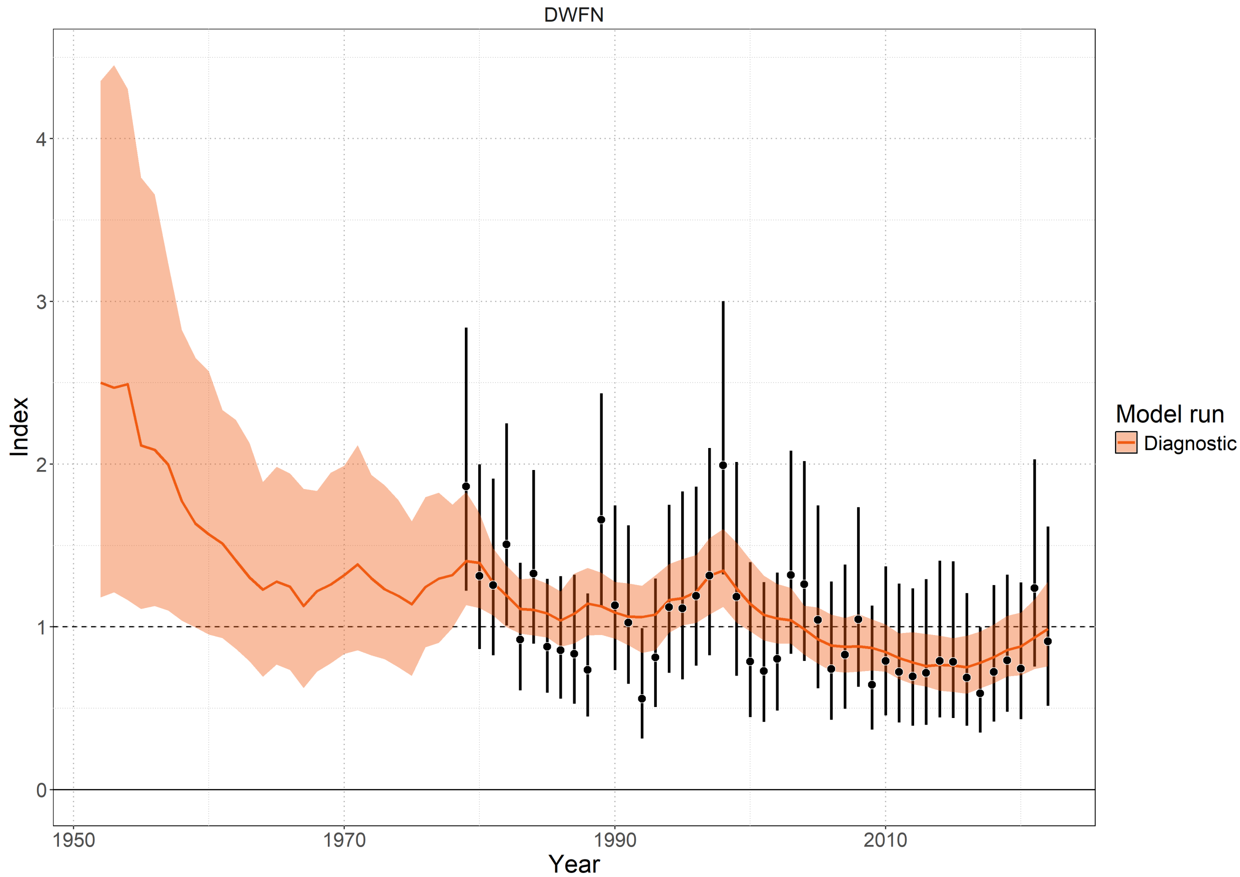

Figure 7: Model fit to the standardized index data showing observations (black points with 95% error bars) and model-predictions (colored lines and shaded ribbons representing 95% credible intervals).

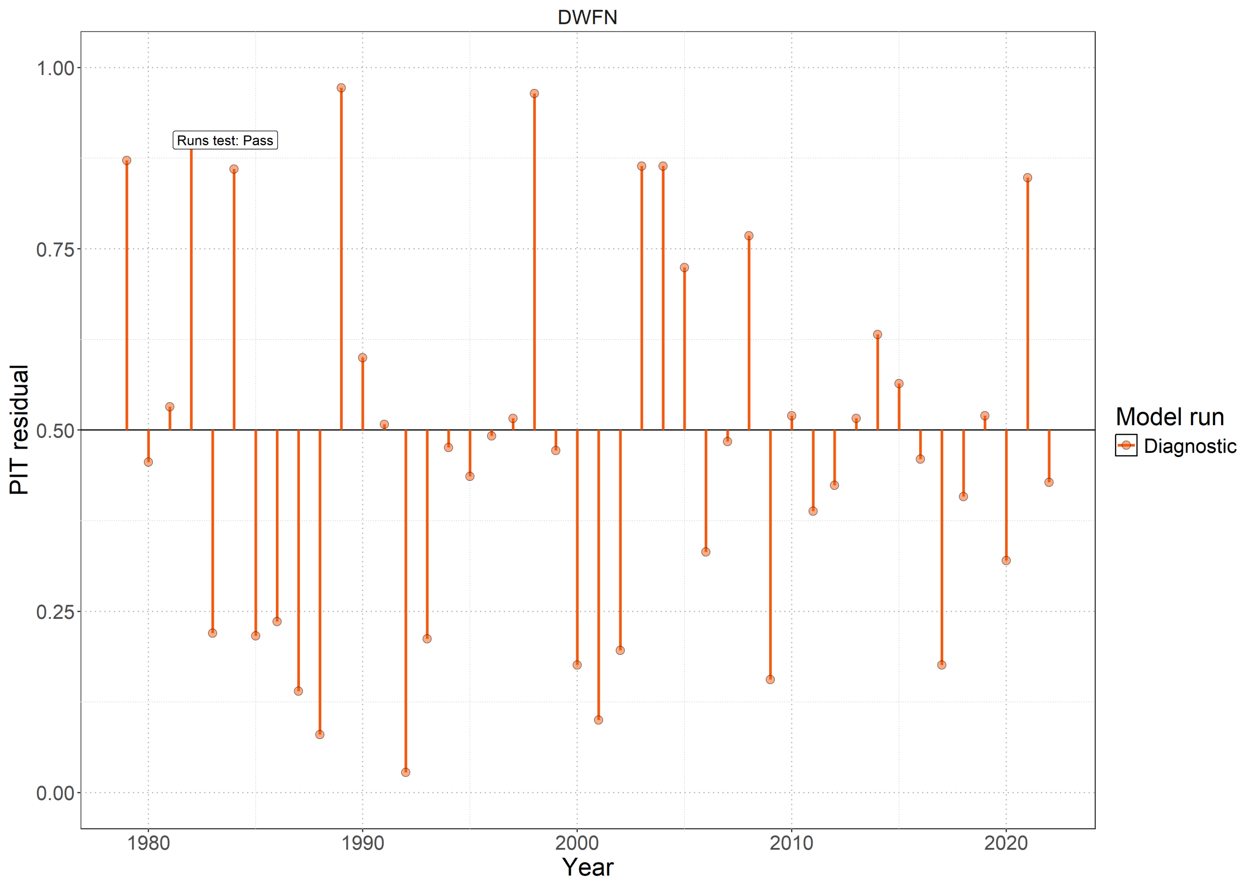

Figure 8: Probability Integral Transform (PIT) residuals for catch data from the diagnostic model over time. Orange points with vertical lines show PIT residual values for each year, where values should be uniformly distributed between 0 and 1 under correct model specification. The 'Runs test: Pass' annotation indicates that the runs test for temporal autocorrelation in the residuals passed at the 5% significance level, suggesting no systematic patterns in the residuals over time.

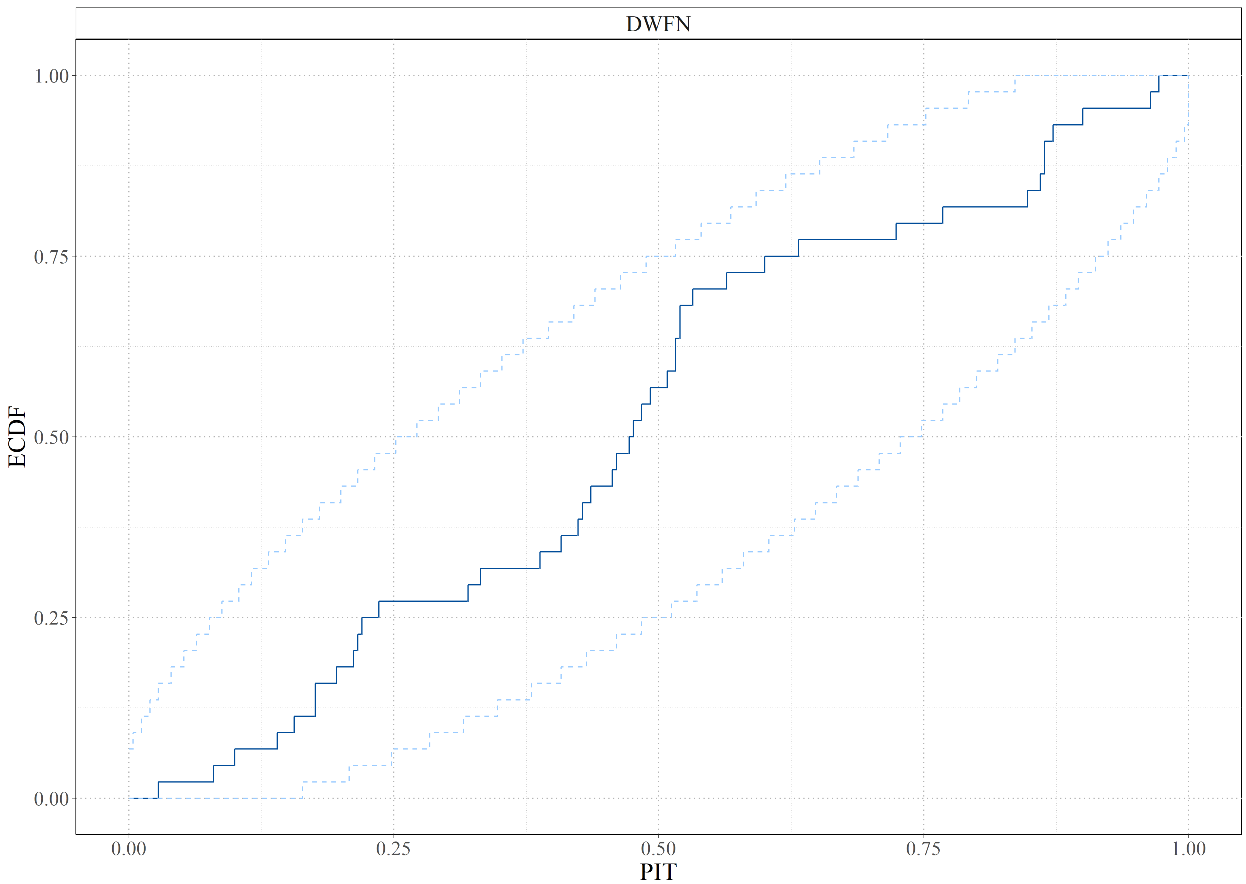

Figure 9: Posterior predictive check for index data using probability integral transform empirical cumulative distribution function (PIT-ECDF). The solid blue line shows the empirical CDF of PIT values calculated from observed catch data relative to posterior predictive distributions. Light blue dashed lines indicate 99% simultaneous confidence bands under the assumption of correct model specification. The solid blue line should be contained within the 99% simultaneous confidence bands.

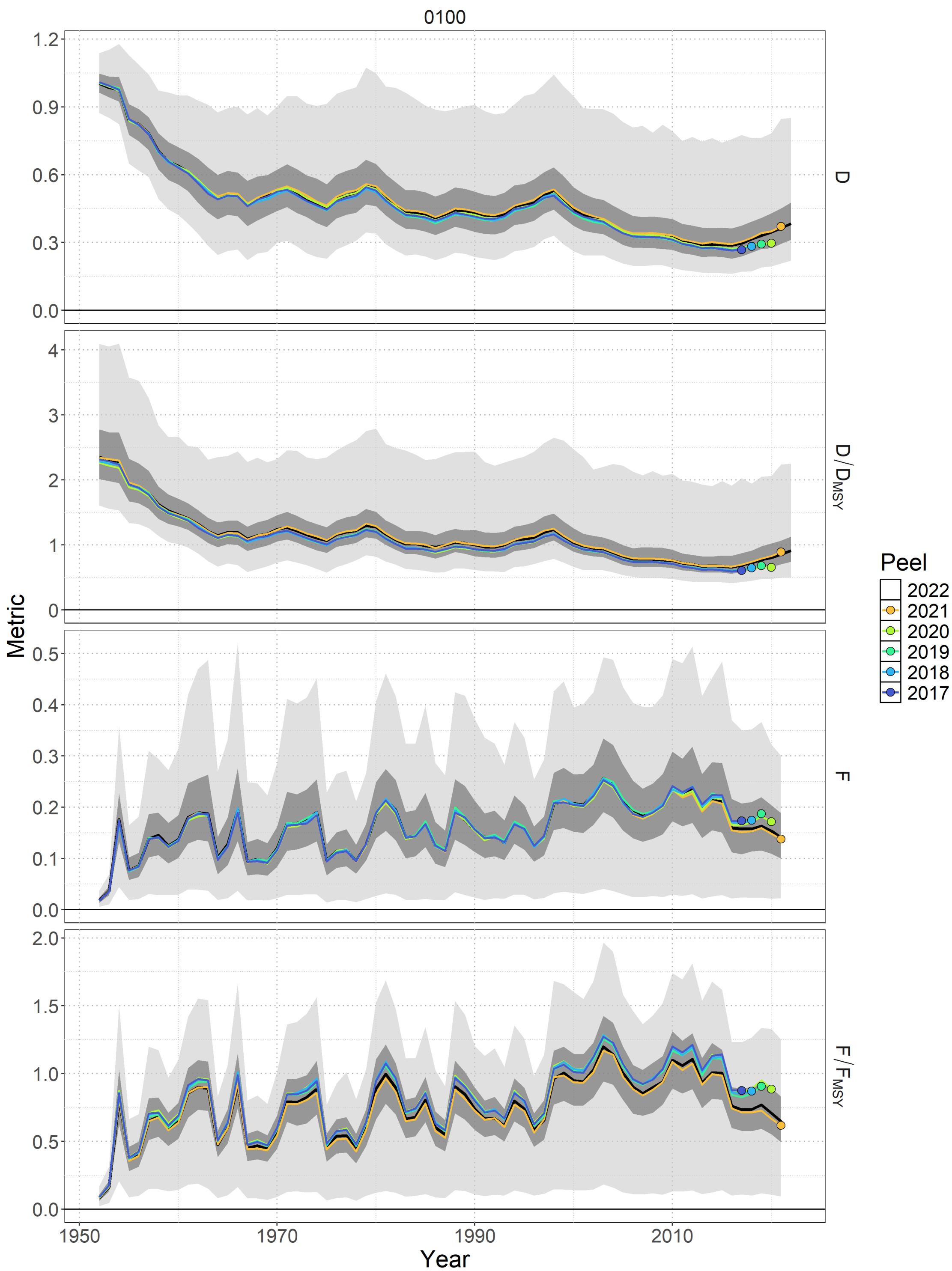

Figure 10: Retrospective analysis for the diagnostic case model (0100) with respect to time series of depletion D_t, depletion relative to depletion at MSY D_t/D_{MSY}, fishing mortality F_t, and fishing mortality relative to the rate of fishing mortality that produces MSY F_t/F_{MSY}. The base model with data included through 2022 (black line – median; dark shading – 50% credible interval; light shading – 95% credible interval) is shown relative to the retrospective models. Colored lines correspond to the last year of index data and the colored point indicates the estimate in the last year of the retrospective peel.

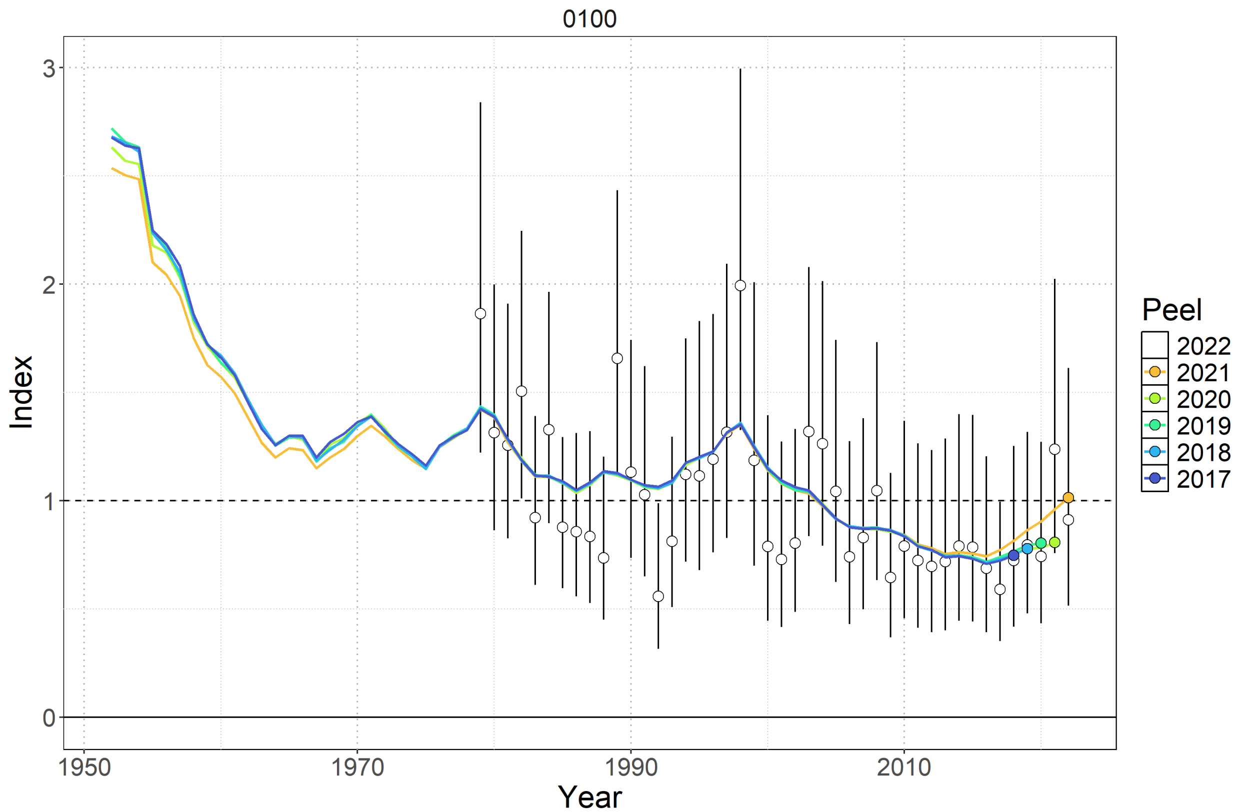

Figure 11: Standardized indices of relative abundance used in the diagnostic case model (`0100`). Open circles show observed values (standardized to mean of 1; black horizontal line) and the vertical bars indicate the observation error (95% confidence interval). One year ahead 'model-free' hindcast predictions are shown by the colored lines, where the color indicates the last year of index data seen by the model. The predicted value is shown one year-ahead with the colored point.

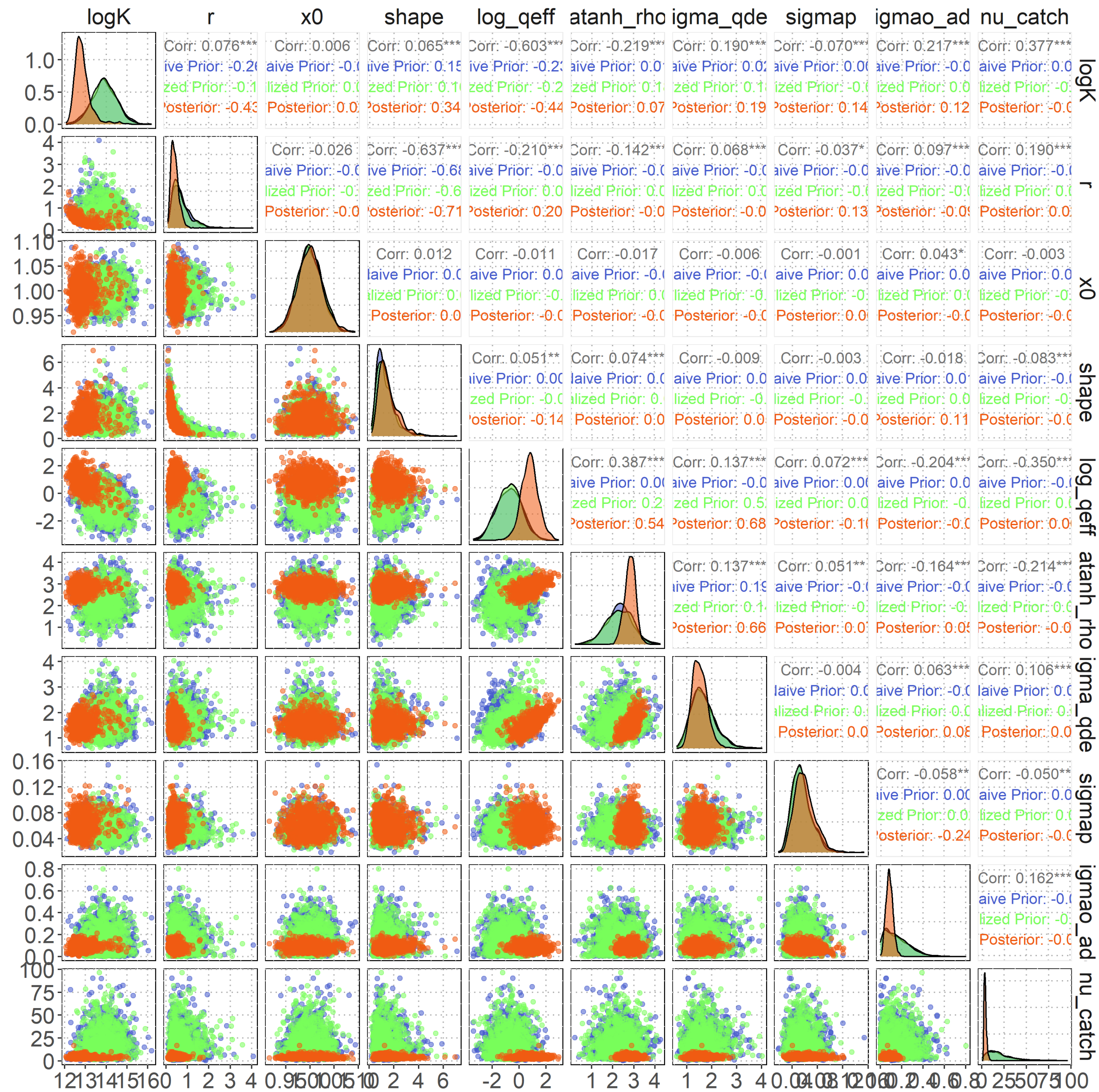

Figure 12: Prior-posterior comparison pairs plot for the diagnostic case model showing naïve prior (blue), realized prior (green), and posterior (orange) distributions across leading model parameters. Diagonal panels display marginal density distributions for each parameter. Off-diagonal scatter plots show pairwise parameter relationships colored by distribution type. Upper triangle correlation coefficients quantify linear relationships between distributions, with statistical significance indicated by asterisks. The comparison illustrates sequential parameter refinement from initial biological/expert knowledge (naïve prior) through model structure filtering (realized prior) to data-based learning (posterior).

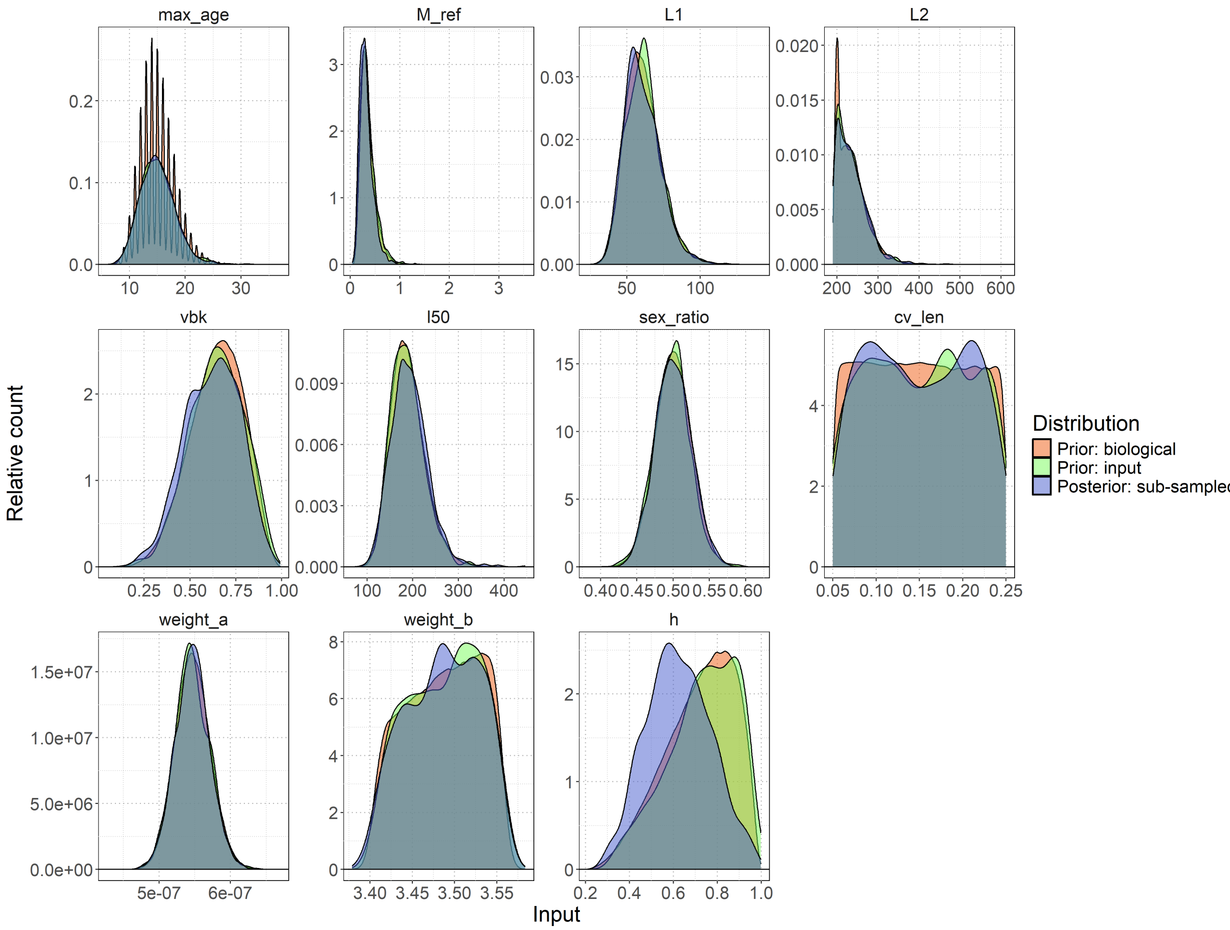

Figure 13: Distributions of biological parameters (maximum age *max_age*, natural mortality reference *M_ref*, length at birth *L1*, asymptotic length *L2*, von Bertalanffy growth coefficient *vbk*, length at 50% maturity *l50*, sex ratio at birth *sex_ratio*, coefficient of variation in length *cv_len*, weight-length relationship parameters *weight_a* and *weight_b*, and steepness *h*) used in the rmax sub-sampling process. Original biological prior distribution (orange shading), input prior distribution following the prior pushforward analysis (green shading), and final sub-sampled distribution selected to match the posterior R_{Max} distribution (blue shading).

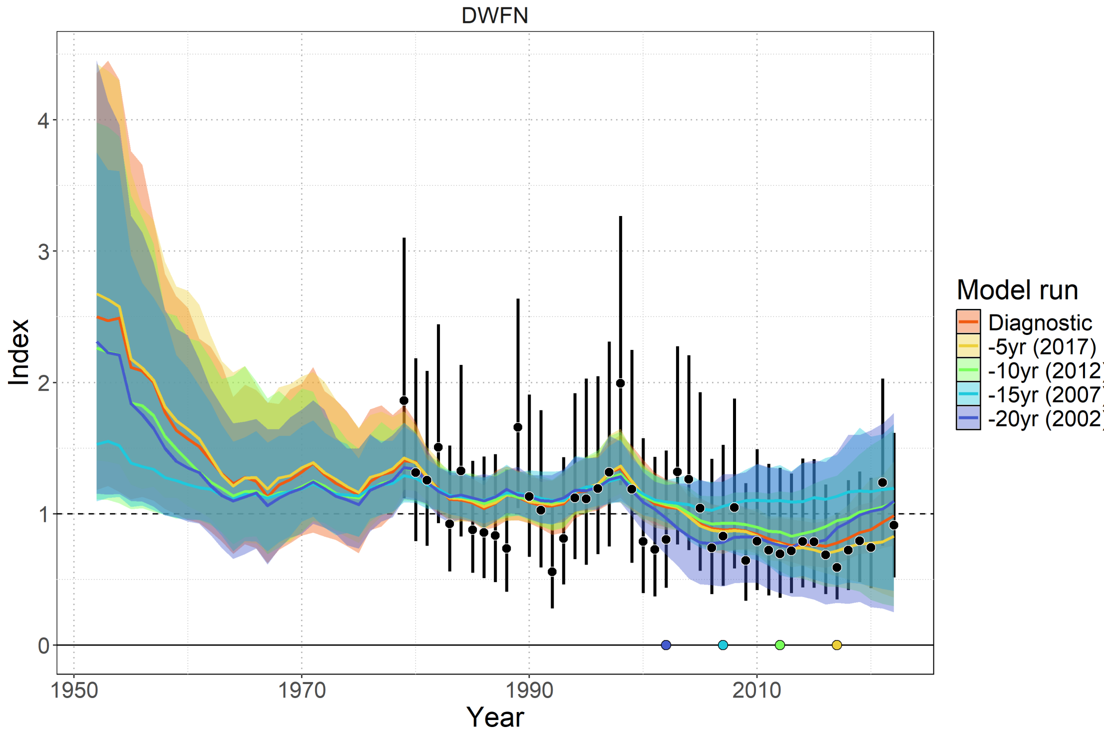

Figure 14: Model-free hindcast performance for the DWFN CPUE index. The diagnostic model fitted to the full dataset (orange line and shading) is compared against retrospective models excluding the final 5, 10, 15, and 20 years of index data. Colored lines show model predictions with 95% credible intervals (shading). Observed index values are shown as black points with error bars. Colored points at the bottom indicate the terminal year of each retrospective peel: 2017 (yellow), 2012 (green), 2007 (cyan), and 2002 (blue).

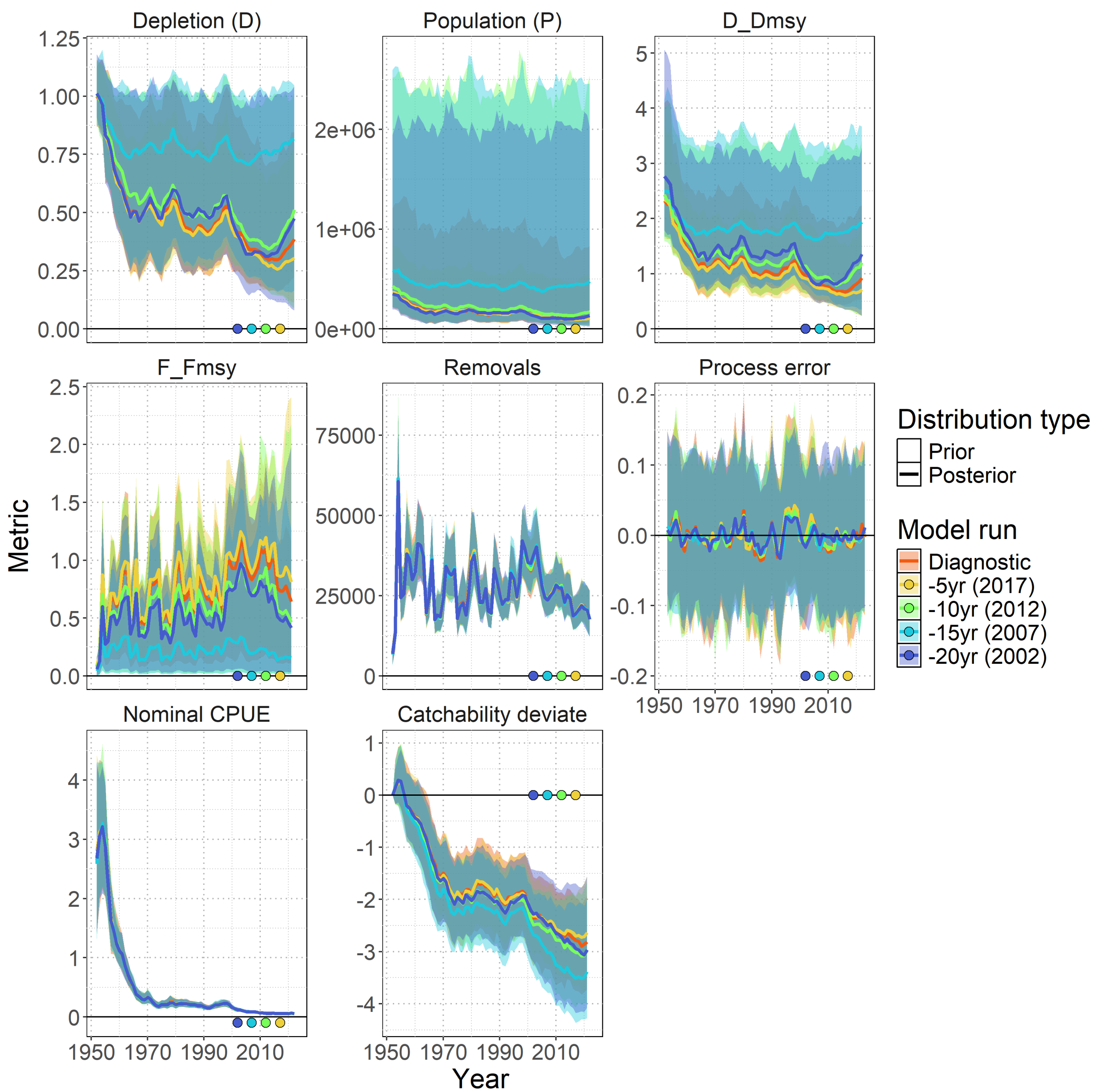

Figure 15: Model-free hindcast performance for key population dynamics parameters. Time series show posterior distributions (median and 95% credible intervals) for depletion (D), population (P), relative depletion (D/D_{MSY}), relative fishing mortality (F/F_{MSY}), removals, process error, nominal CPUE, and catchability deviates. The diagnostic model (orange) is compared against retrospective models terminating in 2017 (yellow), 2012 (green), 2007 (cyan), and 2002 (blue). Colored points indicate the terminal year of each retrospective peel.

11 Technical Annex

Additional technical detail of population dynamics components used to solve the Euler-Lotka equation for R_{Max}.

11.1 Equilibrium Age-Structured Population Dynamics

11.1.1 Survival

Survival schedule was calculated assuming constant natural mortality:

$$l_a = \begin{cases} 1 & \text{if } a = 1 \\ l_{a-1} \times \exp(-M_{ref}) & \text{if } 1 < a < A_{Max} \\ \frac{l_{A_{Max}-1}}{1-\exp(-M_{ref})} & \text{if } a = A_{Max} \text{ (plus group)} \end{cases}$$

11.1.2 Fecundity

Reproductive output incorporated biological constraints:

$$f_a = \frac{\psi_a \times W_a \times s}{\rho}$$

where: - ψ_a = proportion mature at age a - W_a = weight at age a (proxy for reproductive capacity) - s = sex ratio (proportion female) - ρ = reproductive cycle (years between spawning events)

Maturity-at-age was derived from length-based logistic maturity:

$$\psi_L = \frac{\exp(a_{mat} + b_{mat} \times L)}{1 + \exp(a_{mat} + b_{mat} \times L)}$$

where $b_{mat} = -a_{mat}/L_{50}$, then integrated over the length-at-age distribution to obtain ψ_a.

Length-at-age followed the Francis parameterization:

$$L_a = L_1 + (L_2 - L_1) \times \frac{1 - \exp(-k(a - age_1))}{1 - \exp(-k(age_2 - age_1))}$$

with length variability modeled using probability density functions linking length bins to ages.

Weight-at-age used the allometric relationship:

$$W_a = a_w \times L_a^{b_w}$$

11.1.3 Stock-Recruitment Relationship

The slope at the origin of the stock-recruitment curve was:

$$\alpha = \frac{4h}{(1-h)\phi_0}$$

The calculation incorporated steepness (h) through the Beverton-Holt stock-recruitment relationship. Spawners-per-recruit in the unfished state was:

$$\phi_0 = \sum_{a=1}^{A_{Max}} l_a f_a$$

This α parameter links the intrinsic rate of increase to recruitment compensation, ensuring that R_{Max} estimates reflect realistic population productivity under the assumed stock-recruitment dynamics.

Successful simulations (those yielding $R_{Max} > 0$ and $R_{Max} < 1.5$) were filtered to create a plausible prior distribution. The resulting distribution was fitted to a lognormal prior for use in the BSPM.